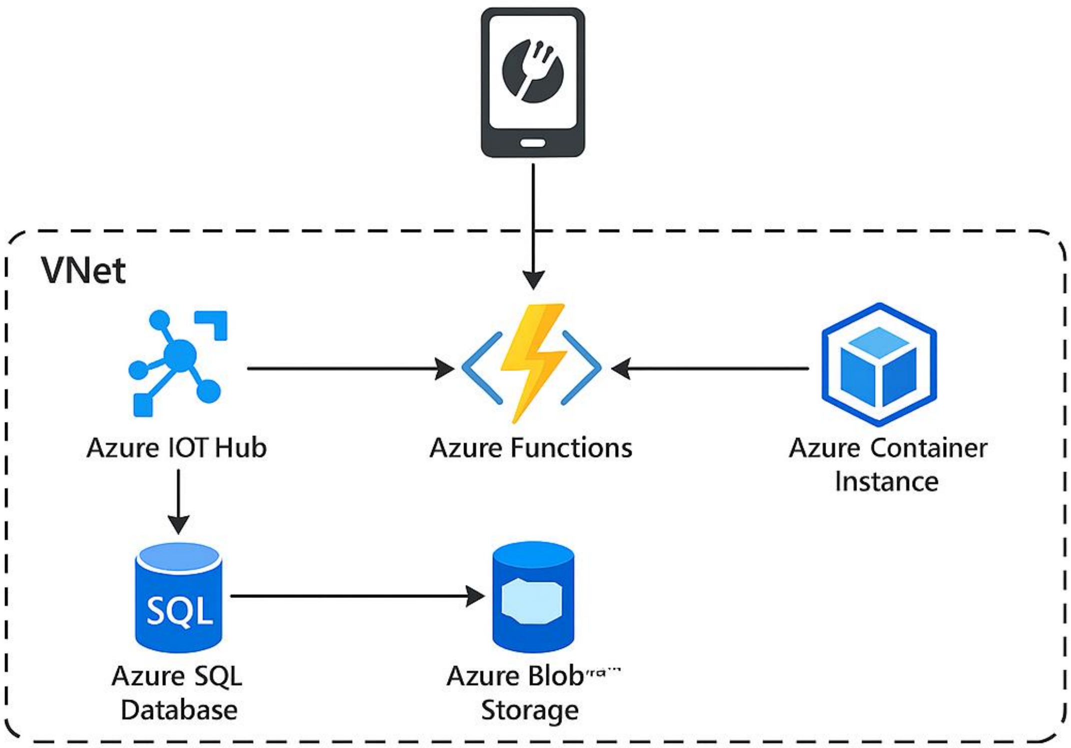
AZ-900 Notes



Service	Description
Azure IoT Hub	Collects real-time order data from devices
Azure SQL Database	Storage and management order-autom information
Azure Functions	Processes order events and business logic
Azure Container Instance	Runs the application's API and business components

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Azure is Always Changing & Updating

Course Links – Expect a Journey – Not a Destination

1. Course AZ-900T00-A: Microsoft Azure Fundamentals – Training | Microsoft Learn
   a. Microsoft Azure Fundamentals: Describe cloud concepts - Training | Microsoft Learn
   b. Microsoft Azure Fundamentals: Describe Azure architecture and services - Training | Microsoft Learn
   c. Microsoft Azure Fundamentals: Describe Azure management and governance - Training | Microsoft Learn
2. Lab Instructions:
   a. kingdomb/AZ-900T0x-Practical_Labs-AzureFundamentals: Microsoft Azure Fundamentals LABS - AZ-900T00 and AZ-900T01
3. Study guide for Exam AZ-900: Microsoft Azure Fundamentals | Microsoft Learn
4. Schedule Exam: Microsoft Certified: Azure Fundamentals - Certifications | Microsoft Learn
5. REFERENCE GUIDE – CHALLENGE LABS: Microsoft Azure FundaMENTALS Collection (AZ-900)

✅ AZ-900 Core Areas & Services

Domain	Key Services / Concepts	Overlap with AZ-204?
Cloud Concepts	IaaS, PaaS, SaaS, Regions, Availability Zones	Yes (AZ-204 uses PaaS heavily)
Core Azure Services	Virtual Machines, App Service, Functions, Storage Accounts	Yes (Compute + Storage)
Core Solutions	IoT Hub, Event Hubs, Azure Synapse, AI services	Yes (Event Hubs, IoT Hub)
Security & Compliance	Azure AD, RBAC, Key Vault, Defender	Yes (Key Vault, RBAC)
Pricing & Support	Cost calculators, SLAs, TCO, Reservations	No direct coding overlap
Monitoring	Azure Monitor, Service Health	Yes (Monitor, Insights)

🔄 Overlap Highlights

- **Storage Accounts**: Both exams require understanding Blob, Queue, Table.
- **Compute**: AZ-204 dives deeper into App Service & Functions (coding), AZ-900 focuses on what they are.
- **Messaging**: Event Hubs & Service Bus appear in both, but AZ-204 emphasizes implementation.
- **Security**: Key Vault and RBAC are common; AZ-204 adds Managed Identity usage.

Treetop

Azure Account, Subscription, Resource Group

Subscription is a billing group

```
Adventureworks sell bycicles

Dept
It - sub
HR - sub
Admin - sub

Business Rule - bill by dept
Subscription - separate billing

Resource Group - Delete a Group as One Operation DevOps
-- Stops Billing Me
----- Recreate it with ARM Template
- Have Many Resources
--- VM, Storage, Vlan,
```

Admin Hierarchies

[Browse Azure Architectures - Azure Architecture Center | Microsoft Learn]

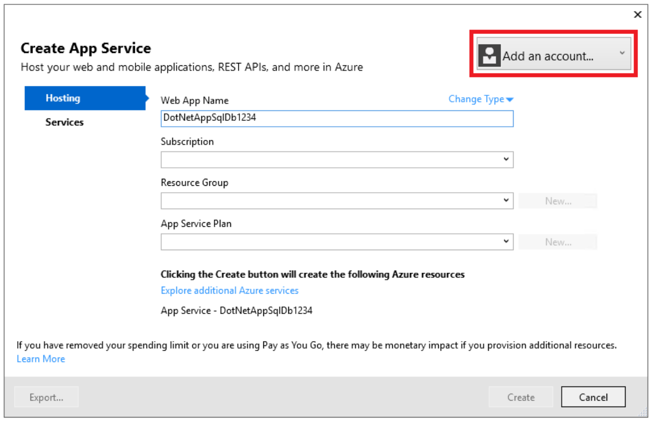
Resource Group

- A resource group is a logical container into which Azure resources like web apps, databases, and storage accounts are deployed and managed together.
- Serves as the lifecycle boundary for every resource that's within it.
- A single resource group can have multiple App Service plans.
- Allocate different apps to different physical resources.
- Multiple plans in a single resource group also define an application that spans geographical regions.
- Example:
  - A highly available app running in two regions includes at least two plans, one for each region, and one app associated with each plan.
  - All the copies of the app are then contained in a single resource group.
  - Having a resource group with multiple plans and multiple apps makes it easy to manage, control, and view the health of the application.

Azure App Service Plan defines a set of compute resources for a web app to run.

1. Azure App Service Plans - Azure App Service | Microsoft Learn
2. Each App Service plan defines:
   a. Region (West US, East US, etc.)
   b. Number of VM instances
   c. Size of VM instances (Small, Medium, Large)
   d. Pricing tier (Free, Shared, Basic, Standard, Premium, PremiumV2, Isolated, Consumption)

3. The pricing tier of an App Service plan determines what App Service features you get and how much you pay for the plan. There are a few categories of pricing tiers:
4. You can save money when hosting multiple apps by configuring the web apps to share a single App Service plan.



Parts List

1. Web App
2. Azure Container Instance
3. Vlan
4. Storage
5. SQL Databases
6. Iot Hub
7. Functions
8. VM
9. Key Vault

Network Security Group

(NSG) in Azure is a resource that you can use to filter network traffic to and from Azure resources within a virtual network.

1. Azure network security groups overview | Microsoft Learn
2. Network security concepts and requirements in Azure | Microsoft Learn

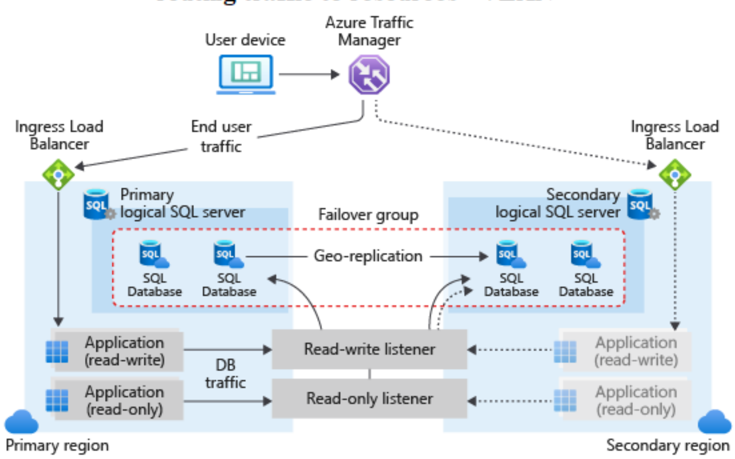
TOC – somewhat legacy

1. Microsoft Azure Fundamentals: Describe Azure architecture and services - Training | Microsoft Learn – top TOC
    a. Describe Azure compute and networking services - Training | Microsoft Learn
        i. Walk tree to find next links -
2. Virtual Desktop – New
    a. To evaluate Azure Virtual Desktop
    b. Deploy Azure Virtual Desktop with the getting started feature | Microsoft Learn

c. *Deliver remote desktops and apps with Azure Virtual Desktop - Training | Microsoft Learn
   i. In depth
d. Describe Azure virtual desktop - Training | Microsoft Learn
e. Deploy Azure Virtual Desktop - Training | Microsoft Learn

3. Mobile App Hosting
   a. About Azure Mobile Apps | Microsoft Learn
   b. Mobile App Service | Microsoft Azure
   c. Introducing a quicker Mobile Apps Quickstart | Azure updates | Microsoft Azure – 2016
   d. Azure Mobile Apps documentation - Azure Mobile Apps | Microsoft Learn
4. WebJob
   a. Run background tasks with WebJobs - Azure App Service | Microsoft Learn
5. Describe Azure virtual networking - Training | Microsoft Learn
   i. Vlan – CTRL-F – IP Addressing for more notes

routing traffic to resources - VLAN

ii. 

iii. JMESPath — JMESPath

```
az network nsg rule list \
   --resource-group [sandbox resource group name] \
   --nsg-name my-vmNSG \
   --query '[].{Name:name, Priority:priority,
   Port:destinationPortRange, Access:access}' \
   --output table
```

iv.

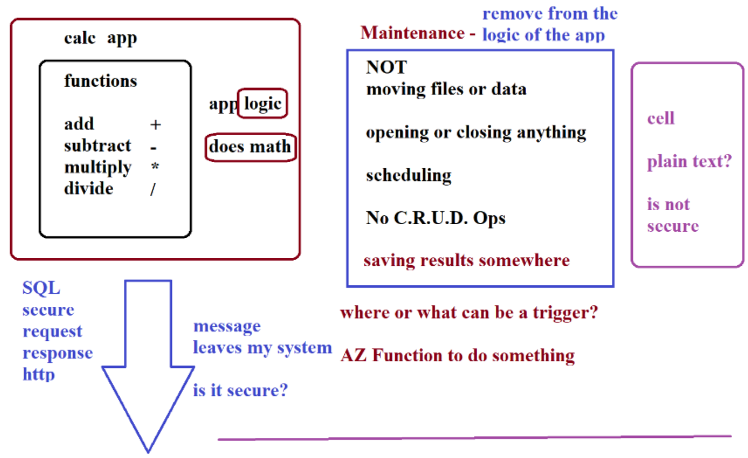
```
locations[?state == 'WA'].name | sort(@) | {WashingtonCities: join(', ', @)}
```

```
{
  "locations": [
    {"name": "Seattle", "state": "WA"},
    {"name": "New York", "state": "NY"},
    {"name": "Bellevue", "state": "WA"},
    {"name": "Olympia", "state": "WA"}
  ]
}
```

JMESPath is a query language for JSON

Result

```
{
  "WashingtonCities": "Bellevue, Olympia, Seattle"
}
```



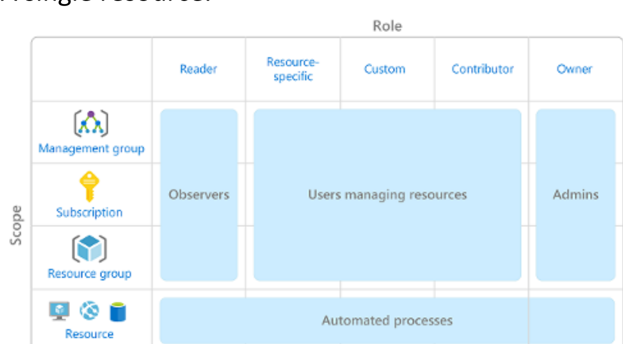
v.
vi.

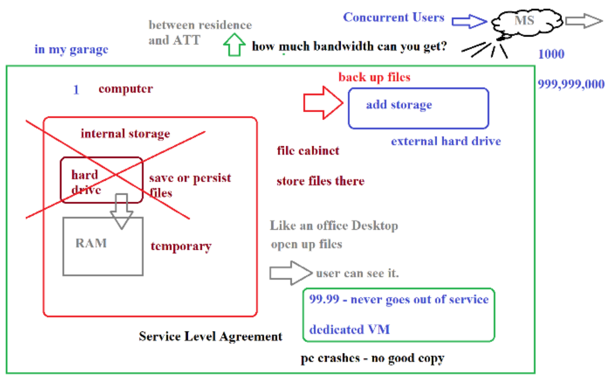
Identity, Access, and Security ppt02-48

For on-premises environments, Active Directory running on Windows Server provides an identity and access management service that's managed by your organization. Microsoft Entra ID is Microsoft's cloud-based identity and access management service.

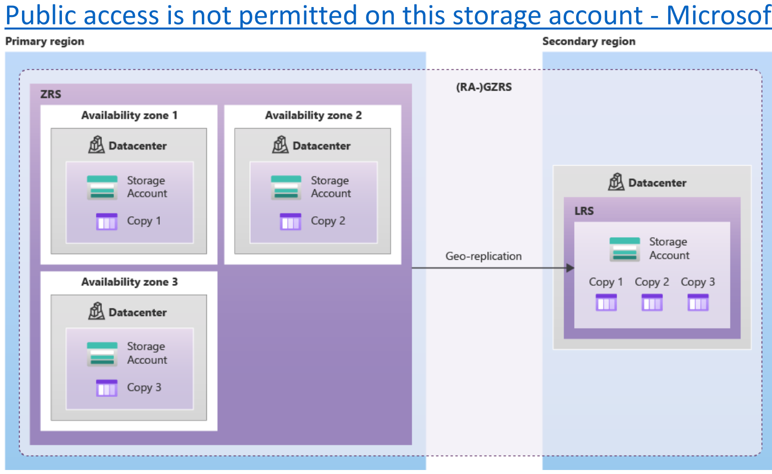
When you secure identities on-premises with Active Directory, Microsoft doesn't monitor sign-in attempts. When you connect Active Directory with Microsoft Entra ID, Microsoft can help protect you by detecting suspicious sign-in attempts at no extra cost.

Microsoft Entra ID is a directory service

6. Describe Azure directory services - Training | Microsoft Learn
7. Describe Azure identity, access, and security - Training | Microsoft Learn
8. Describe Azure role-based access control - Training | Microsoft Learn:
    a. Azure RBAC group: Role-based access control is applied to a scope
    b. A management group (a collection of multiple subscriptions).
    c. A single subscription.
    d. A resource group.
    e. A single resource.
    f. 

g. 

Azure Storage ppt02-sl37

9. [Introduction to Azure Storage - Cloud storage on Azure | Microsoft Learn]
10. [Describe Azure storage services - Training | Microsoft Learn] - #12 in labs
    a. [Azure Blobs]: A massively scalable object store for text and binary data. Also includes support for big data analytics through Data Lake Storage.
    b. [Azure Files]: Managed file shares for cloud or on-premises deployments.
    c. [Azure Elastic SAN]: A fully integrated solution that simplifies deploying, scaling, managing, and configuring a SAN in Azure.
    d. [Azure Queues]: A messaging store for reliable messaging between application components.
    e. [Azure Tables]: A NoSQL store for schemaless storage of structured data. JSON
    f. [Azure managed Disks]: Block-level storage volumes for Azure VMs.
    g. [Azure Container Storage]: A volume management, deployment, and orchestration service built natively for containers.
    h. Static web site storage
    i. [Quickstart: Upload, download, and list blobs - Azure portal - Azure Storage | Microsoft Learn]
    j. Static site for fun:
    k. https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blob-static-website
    l. [Public access is not permitted on this storage account - Microsoft Community Hub]
    m. 

11. [Microsoft Azure Fundamentals: Describe Azure management and governance - Training | Microsoft Learn]
    a. Resource locks, Purview, Service Trust portal etc.. ppt 03

12. Manage resource locks
    a. https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/lock-resources?tabs=json

Azure Migrate

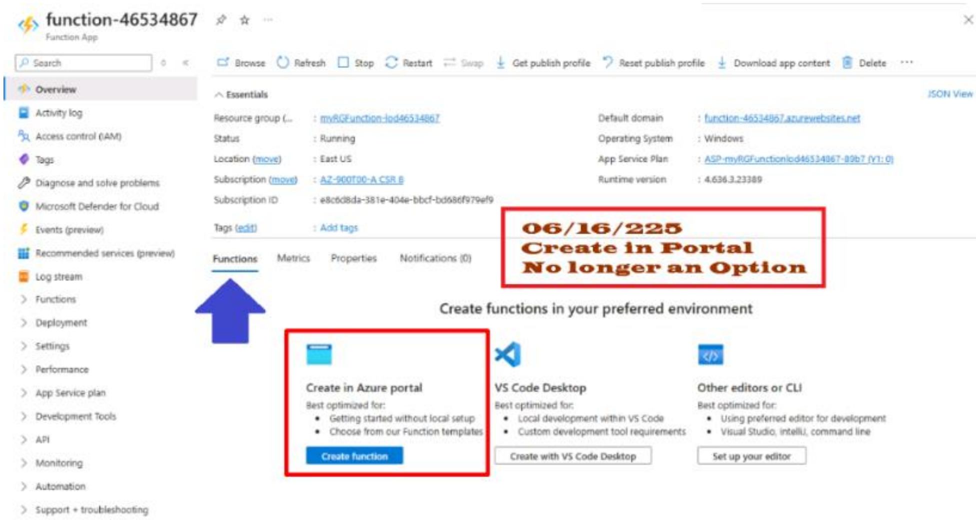
1. Identify Azure data migration options - Training | Microsoft Learn
2. Azure Migrate: Discovery and assessment. Discover and assess on-premises servers running on VMware, Hyper-V, and physical servers in preparation for migration to Azure.
3. Azure Migrate: Server Migration. Migrate VMware VMs, Hyper-V VMs, physical servers, other virtualized servers, and public cloud VMs to Azure.
4. Data Migration Assistant. Data Migration Assistant is a stand-alone tool to assess SQL Servers. It helps pinpoint potential problems blocking migration. It identifies unsupported features, new features that can benefit you after migration, and the right path for database migration.
5. Azure Database Migration Service. Migrate on-premises databases to Azure VMs running SQL Server, Azure SQL Database, or SQL Managed Instances.
6. Azure App Service migration assistant. Azure App Service migration assistant is a standalone tool to assess on-premises websites for migration to Azure App Service. Use Migration Assistant to migrate .NET and PHP web apps to Azure.
7. Azure Data Box. Use Azure Data Box products to move large amounts of offline data to Azure.
8. Identify Azure file movement options - Training | Microsoft Learn – AzCopy, Azure Storage Explorer, Azure File Sync

TOC – from intro notes doc that goes with extreme labs...legacy

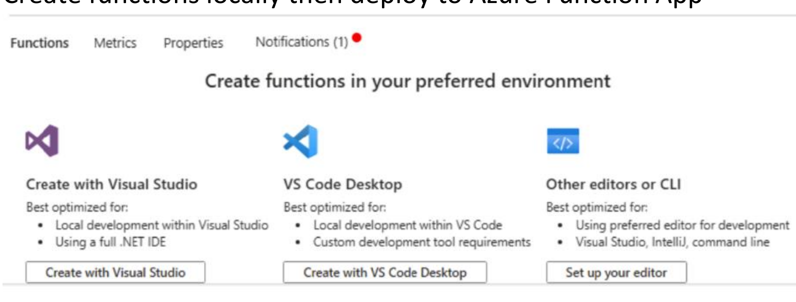
13. Describe features and tools for managing and deploying Azure resources - Training | Microsoft Learn
14. Azure Arc | Microsoft Learn
15. Azure Arc allows you to manage resource types hosted outside of Azure:
    a. Servers, Kubernetes clusters, Azure data services, SQL Server, Virtual machines
    b. Quickstart: Connect an existing Kubernetes cluster to Azure Arc - Azure Arc | Microsoft Learn
    c. Overview | Azure Arc Jumpstart - "zero to hero" experience
16. Describe monitoring tools in Azure - Training | Microsoft Learn
17. Quickstart - Deploy Docker container to container instance - Portal - Azure Container Instances | Microsoft Learn
18. Overview of Azure Blueprints - Azure Blueprints | Microsoft Learn
19. Get started with Azure Monitor metrics explorer - Azure Monitor | Microsoft Learn
20. Sources of data in Azure Monitor - Azure Monitor | Microsoft Learn
21. Pay As You Go—Buy Directly | Microsoft Azure
22. End list

Azure Functions – Scale Scaling Hosting options

1. [Create a function in Azure from the command line | Microsoft Learn]()
    a. Create CMD line locally then deploy inside lab
2. [Create an HTTP endpoint using Azure Functions in the portal | Microsoft Learn]()
3. Function Lab Update – do Local version in Skillable as if 06/2025
4. Azure Functions must be created locally then Deployed to Function App – no more Templates in the Portal.



    a.
    b. Create functions locally then deploy to Azure Function App



    c.
5. [Getting started with Azure Functions | Microsoft Learn]()
6. [Create a Python function using Visual Studio Code - Azure Functions | Microsoft Learn]()
7. [Create a function app in the Azure portal | Microsoft Learn]()
    a. You can now deploy a code project to the function app resources you created in Azure.
    b. [Create a function in Azure from the command line | Microsoft Learn]()
8. [Develop Azure Functions by using Visual Studio Code | Microsoft Learn]()
9. [Work with access keys in Azure Functions | Microsoft Learn]()
    a. In Functions, access keys are randomly generated 32-byte arrays that are encoded as URL-safe base-64 strings
10. [Azure Function app and an HTTP-triggered function - Code Samples | Microsoft Learn]()
11. [Execute an Azure Function with triggers - Training | Microsoft Learn]()
12. [Azure Functions Dedicated hosting | Microsoft Learn]()
13. [Azure Functions scale and hosting | Microsoft Learn]()
    a. In order to use the Monitoring Services outlined in the last step of this lab, please follow these instructions to set up a Log Analytics Workspace.

14. App Service Editor · projectkudu/kudu Wiki – Visual Studio Online (Manaco)

Hubs Iot Event Messaging

1. Create an Azure IoT hub - Azure IoT Hub | Microsoft Learn
    a. The Internet of Things (IoT) is a network of physical devices that connect to and exchange data with other devices and services over the Internet or other network.
    b. Azure IoT Hub provides the capability to stream data from your connected devices and integrate that data into your business applications.
    c. Azure Event Grid is a fully managed event routing service that uses a publish-subscribe model. IoT Hub and Event Grid work together to integrate IoT Hub events into Azure and non-Azure services, in near-real time. IoT Hub publishes both device events and telemetry events.

2. Tutorial - Configure message routing - Azure IoT Hub | Microsoft Learn
3. Compare Event Grid, routing for IoT Hub - Azure IoT Hub | Microsoft Learn
4. Azure Event Grid to enable your business to react quickly to critical events.
5. Azure Logic Apps to automate business processes.
6. Azure Machine Learning to add machine learning and AI models to your solution.
7. Azure Stream Analytics to run real-time analytic computations on the data streaming from your devices.

Azure IoT Edge

Azure IoT Edge brings edge-based capabilities to a cloud-based solution and is a feature of Azure IoT Hub that enables you to scale out and manage an IoT solution from the cloud.

It's difficult to manage the software lifecycle for millions of IoT devices that are often different makes and models or geographically scattered.

Workloads are created and configured for a particular type of device, deployed to all of your devices, and monitored to catch any misbehaving devices. These activities can't be done on a per-device basis and must be done at scale.

Azure IoT Edge is made up of three components:

- **IoT Edge modules** are containers that run Azure services, third-party services, or your own code. Modules are deployed to IoT Edge devices and execute locally on those devices.
- The **IoT Edge runtime** runs on each IoT Edge device and manages the modules deployed to each device.
- A **cloud-based interface** enables you to remotely monitor and manage IoT Edge devices.

By packaging your business logic into standard containers and using optional pre-built IoT Edge module images from partners or the Microsoft Artifact Registry, you can easily compose, deploy, and maintain your solution.

IoT Edge Devices are internet-connected devices with their own processing power that collect and analyze data at the source, like a smart security camera or a factory sensor.

This "edge computing" approach allows for real-time decision-making, reduced bandwidth usage, and improved security by processing data locally instead of sending everything to a central cloud.

Examples range from consumer devices like smart TVs to industrial sensors and even drones used for inspections.

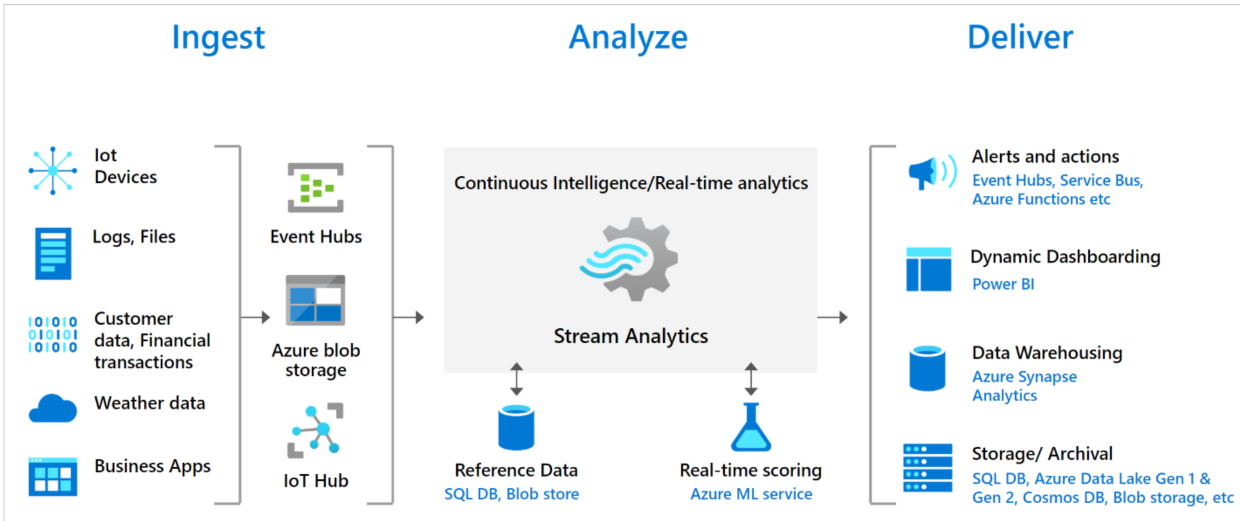
IoT Edge Modules are units of execution, implemented as Docker-compatible containers, that run your business logic at the edge. Multiple modules can be configured to communicate with each other, creating a pipeline of data processing. You can develop custom modules or package certain Azure services into modules that provide insights offline and at the edge.

https://learn.microsoft.com/en-us/azure/iot-edge/about-iot-edge

Azure Stream Analytics

1. Azure Stream Analytics is a fully managed stream processing engine that is designed to analyze and process large volumes of streaming data with sub-millisecond latencies.
2. It allows you to write SQL-like queries to transform and analyze live data from sources like Event Hubs or IoT Hub, and output the results to various other services such as Power BI, Azure SQL, or Azure Blob Storage.

3. You can build a streaming data pipeline using Stream Analytics to identify patterns and relationships in data that originates from various input sources including applications, devices, sensors, clickstreams, and social media feeds.
4. Then, you can use these patterns to trigger actions and initiate workflows such as raising alerts, feeding information to a reporting tool, or storing transformed data for later use.
5. Stream Analytics is also available on the Azure IoT Edge runtime, which enables you to process data directly from IoT devices.
6. Key features include built-in temporal logic, anomaly detection, and integration with Azure Machine Learning for predictive scoring.



Key capabilities

- **Real-time processing:**

Analyzes streaming data in real-time, with sub-millisecond latencies.

- **SQL-like query language:**

Uses a declarative SQL-like language for defining transformations and analyses.

- **Native integrations:**

Integrates with numerous Azure services, including Event Hubs, IoT Hub, Azure Blob Storage, and Power BI, without complex setup.

- **Built-in temporal operators:**

Supports temporal logic for tasks like windowed aggregates, temporal joins, and temporal analytic functions.

- **Anomaly detection:**

Includes built-in functions for anomaly detection in streaming data.

- **Geospatial analytics:**

Supports geospatial functions for use cases like fleet management and geofencing.

- **Custom code:**

Allows for custom logic extensions through callouts to JavaScript user-defined functions.

- **Predictive scoring:**

Can integrate with Azure Machine Learning to score streaming data in real-time.

Common use cases

- Analyzing real-time telemetry from IoT devices.
- Performing clickstream analytics to understand user behavior.
- Implementing anomaly detection in sensor data.
- Monitoring high-value assets and performing predictive maintenance.
- Analyzing real-time logs from applications.

Getting started

1. **Sign in to the Azure portal**: and navigate to your Event Hubs or IoT Hub, which contains your incoming data.
2. **Process data:** Select Process data from the left menu.
3. **Use the no-code editor or test a query:** Choose to start with a template in the no-code editor or write a query in the query page.
4. **Create a Stream Analytics job:** Once you've tested your query, select Create Stream Analytics job.
5. **Configure input and output:** Provide a name for the job and add the input (your Event Hub/IoT Hub) and an output of your choice (e.g., Power BI, Azure SQL).
6. **Start the job:** Save your query and start the job to begin processing data in real-time.

- Introduction to Azure Stream Analytics - Azure Stream Analytics | Microsoft Learn
- Send device telemetry to Azure IoT Hub tutorial - Azure IoT | Microsoft Learn

Discussion Diagrams

login sql database

climbing a ladder

what are the steps to anything

fill out the form

parent/child

sub
RG
Resources

Adventureworks - Bicycle Co. - Organization - Grouping (Container Discussion) - Dept - IT - HR - Admin

Service API - App~~lication~~ Programming Interface

UI/UX
Cell
Phone

http://myDomain.com
https:// - SSL Secure Socket Layer - private to your device

Azure
SQL App
Tims Eats
server side

pipeline - connection

JSON - name:value pairs - querystring - message

concurrent users - number of users same time
can the machines handle?

columns:rows - firstName=Tim

client side

My App - Framework

DevOps Pipeline

design build test approve deploy

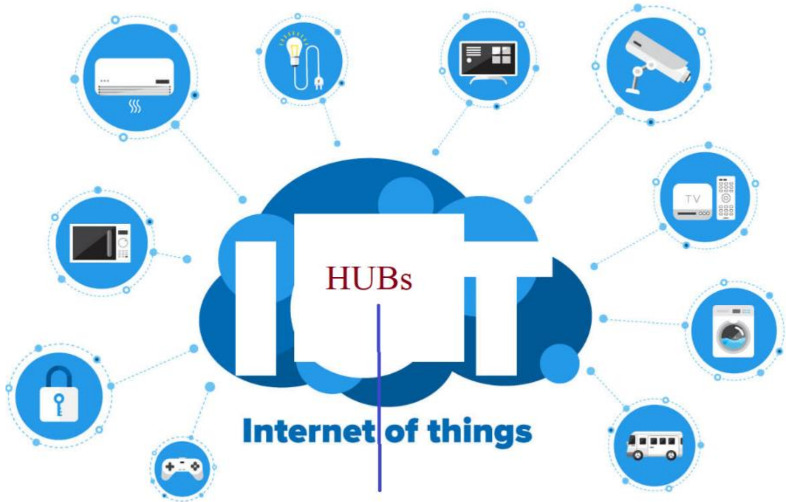
describe workflow steps - lifecycle - automate the process with JSON Arm Templates

Office 365 - Word, Excel - Active Directory - SQL Database

Power BI Service

Azure Account

Active Directory



Features

Software point of view - What can you do?

Virtual Machine - 1 resource

operating system Windows data - 2019

billing - cost?

hard drive

ssd - hhd

motherboards connects things

CPUs

RAM memory

vm - web server
IIS - Internet Information Services
Linux - Apache

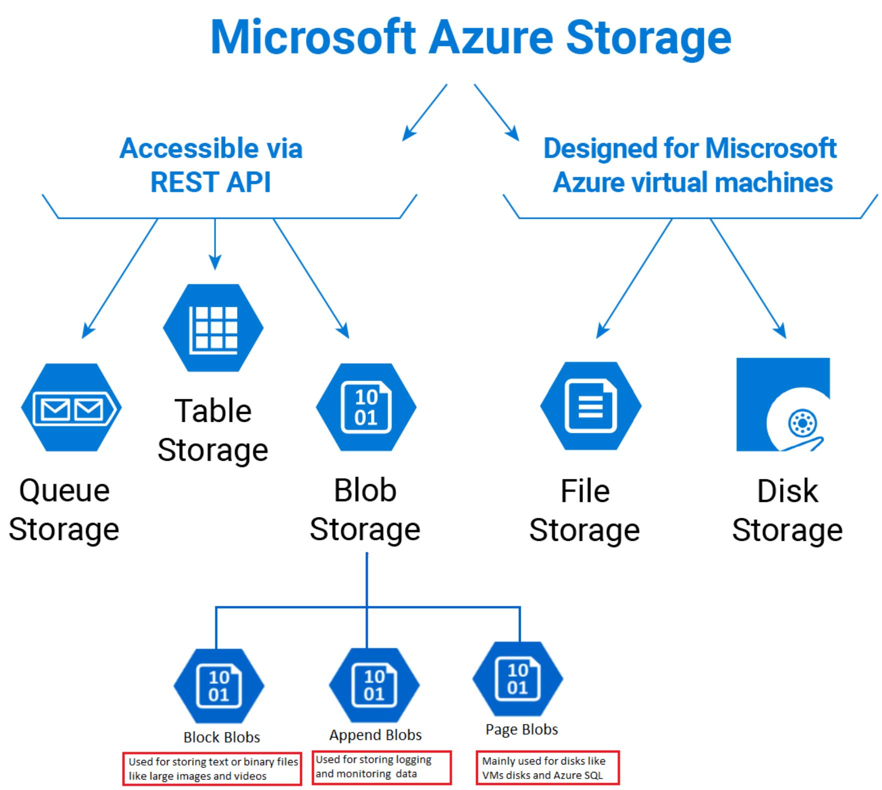
map file location to domain and ip address

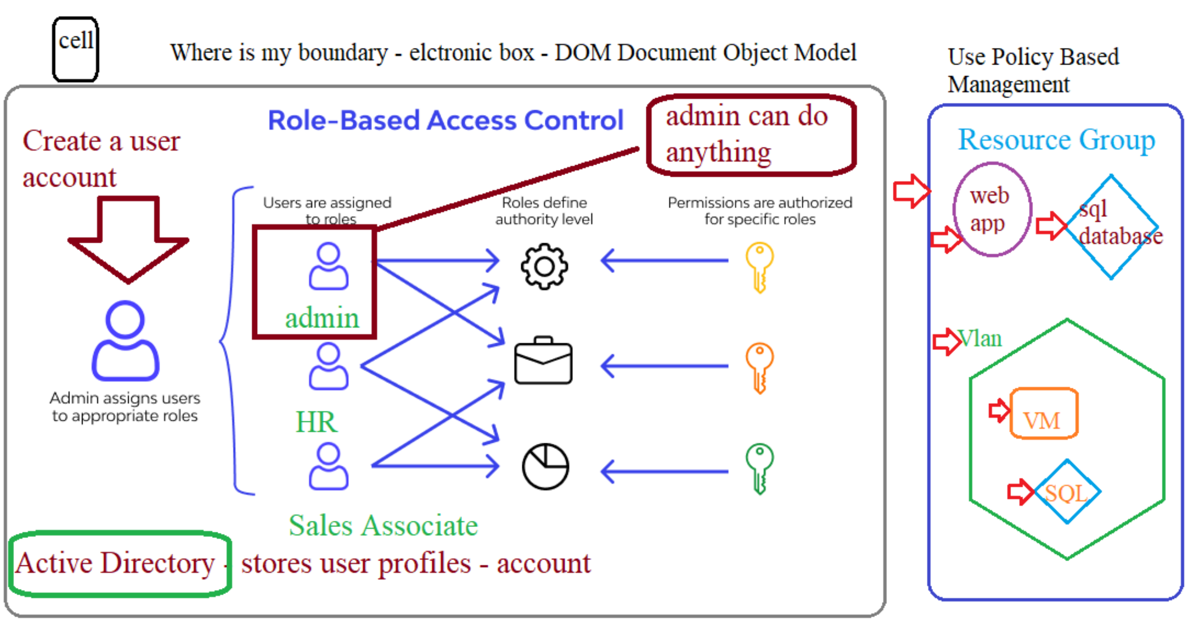
c:\inetpub\wwwroot\myWebSite\home.html

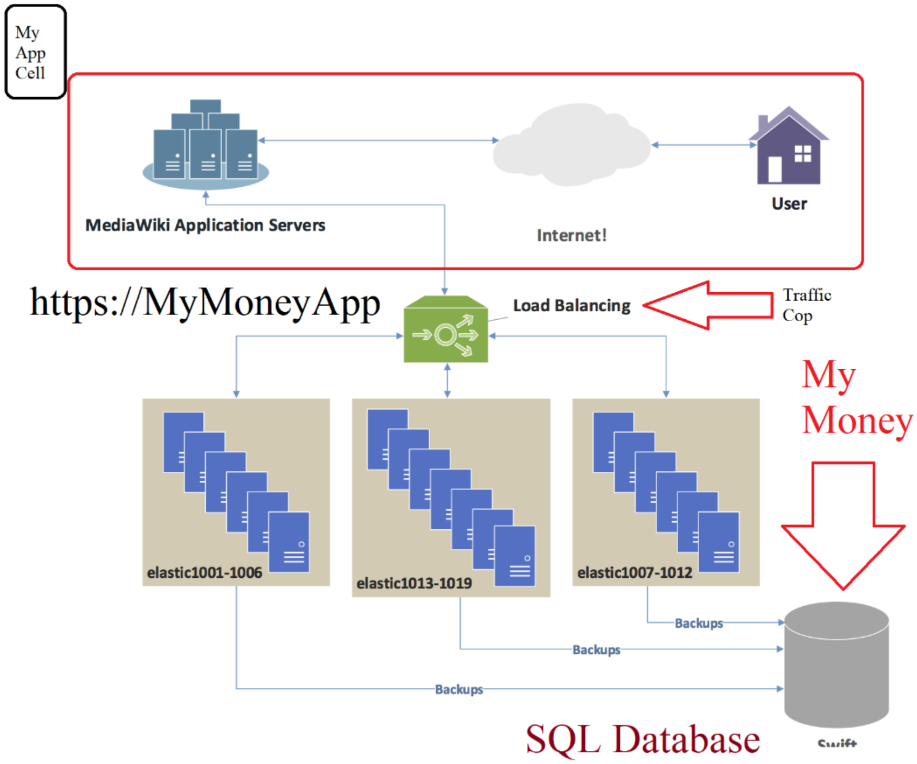
in a browser using http

http://myDomain.com

ip address: 192.168.1.101


Microsoft Azure Storage
Accessible via REST API
Designed for Miscrosoft Azure virtual machines
Queue Storage
Table Storage
Blob Storage
File Storage
Disk Storage
10 01
Block Blobs
Used for storing text or binary files like large images and videos
Append Blobs
Used for storing logging and monitoring data
Page Blobs
Mainly used for disks like VMs disks and Azure SQL


cell
Where is my boundary - elctronic box - DOM Document Object Model
Use Policy Based Management
Role-Based Access Control
admin can do anything
Create a user account
Admin assigns users to appropriate roles
Users are assigned to roles
Roles define authority level
Permissions are authorized for specific roles
admin
HR
Sales Associate
Active Directory - stores user profiles - account
Resource Group
web app
sql database
Vlan
VM
SQL

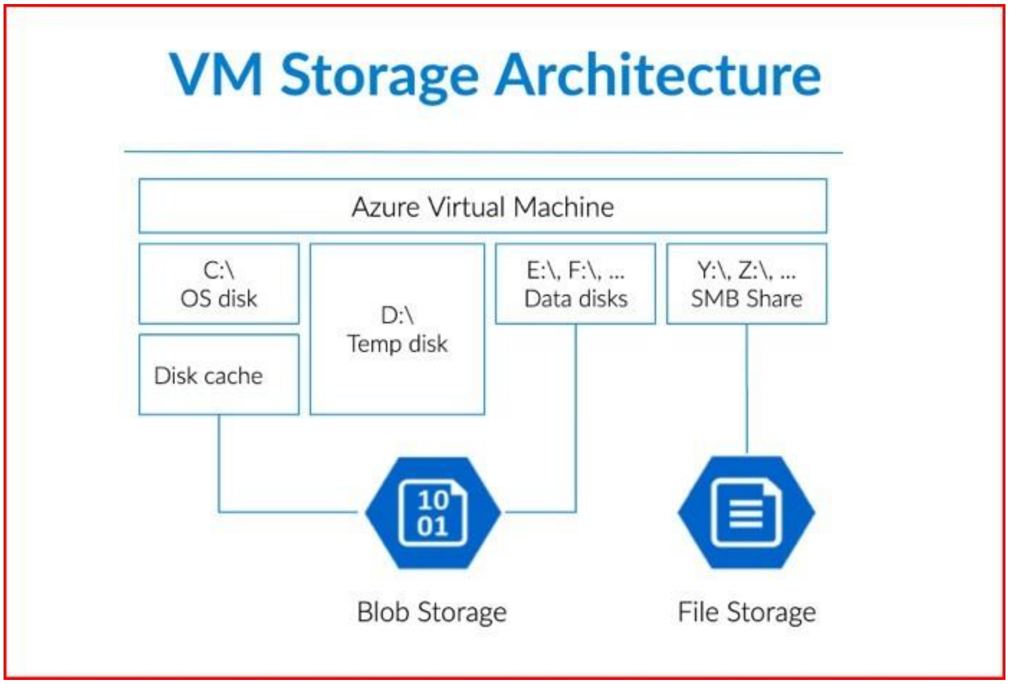


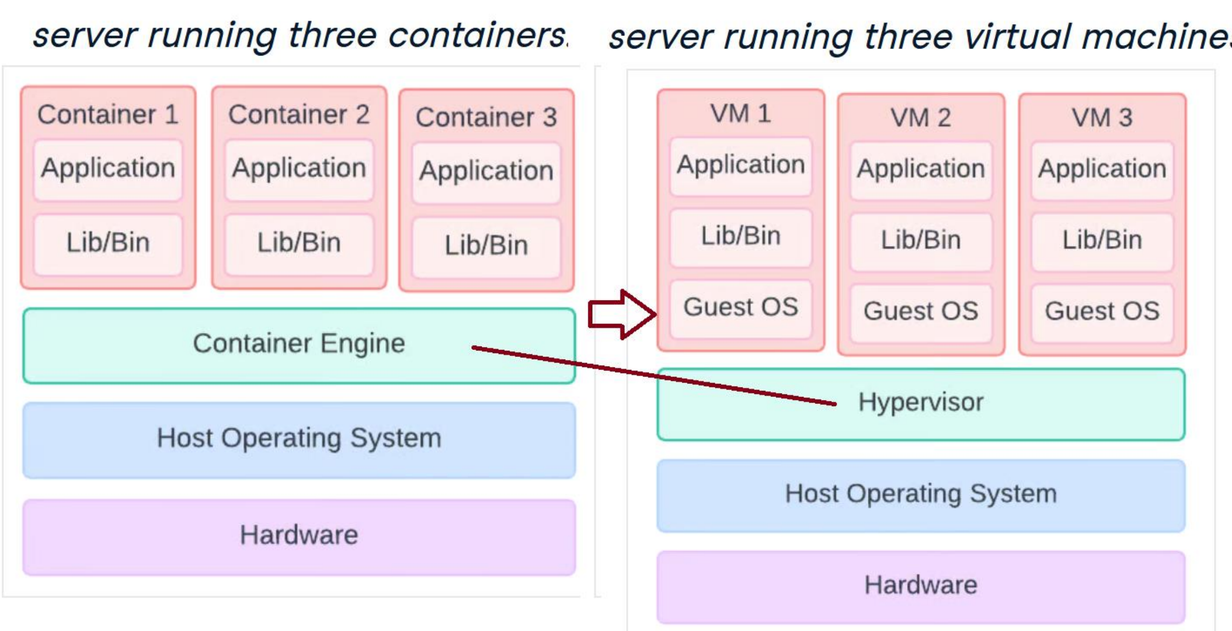
[Azure information system components and boundaries | Microsoft Learn]

Virtual Machines & Parts that go with it (ppt 02 sl23) Azure Recovery Azure Backup

When you create a virtual machine, you're also creating resources that support the virtual machine. These resources come with their own costs that should be considered.

1. Overview of virtual machines in Azure - Azure Virtual Machines | Microsoft Learn
    a. Managed Disks and Storage
2. Create a Windows virtual machine in Azure - Training | Microsoft Learn
3. Create a nested virtual machine in a Microsoft Azure Linux VM | Open-Source Routing and Network Simulation (brianlinkletter.com)
4. Describe Azure virtual machines - Training | Microsoft Learn
5. Availability options for Azure Virtual Machines - Azure Virtual Machines | Microsoft Learn
6. Overview backup options for VMs - Azure Virtual Machines | Microsoft Learn
7. Attach a managed data disk to a Windows VM - Azure - Azure Virtual Machines | Microsoft Learn
8. Monitor Azure Virtual Machines - Azure Virtual Machines | Microsoft Learn
9. What's new in Azure Disk Storage - Azure Virtual Machines | Microsoft Learn
10. Containers vs. virtual machines | Microsoft Learn – aka Docker
11. Microsoft | Docker Hub
12. Microsoft Artifact Registry

13. 



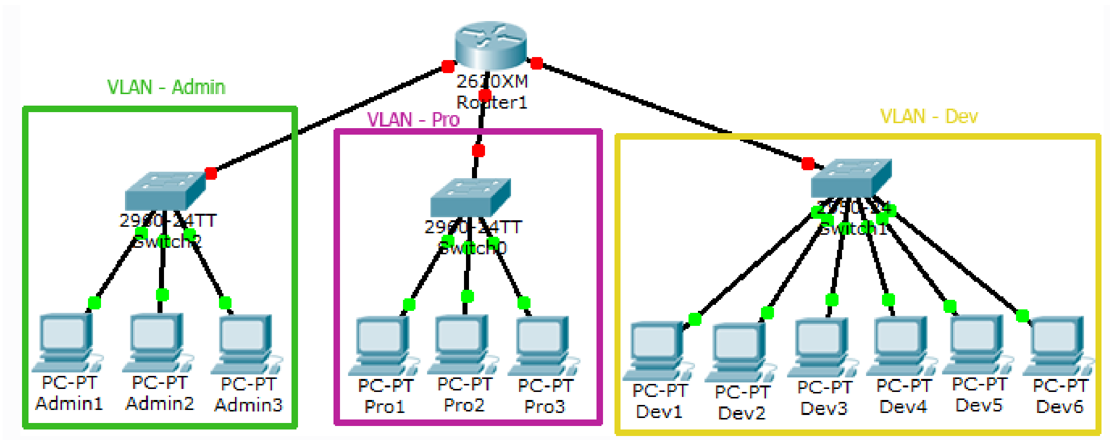
Azure Elastic SAN

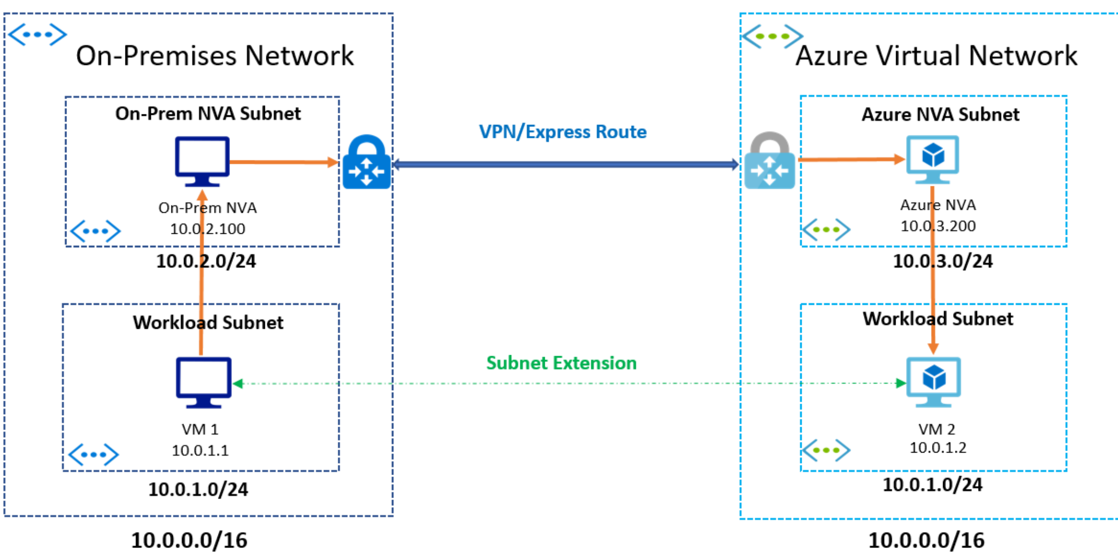
Consolidate the storage for all your workloads into a single storage back end.

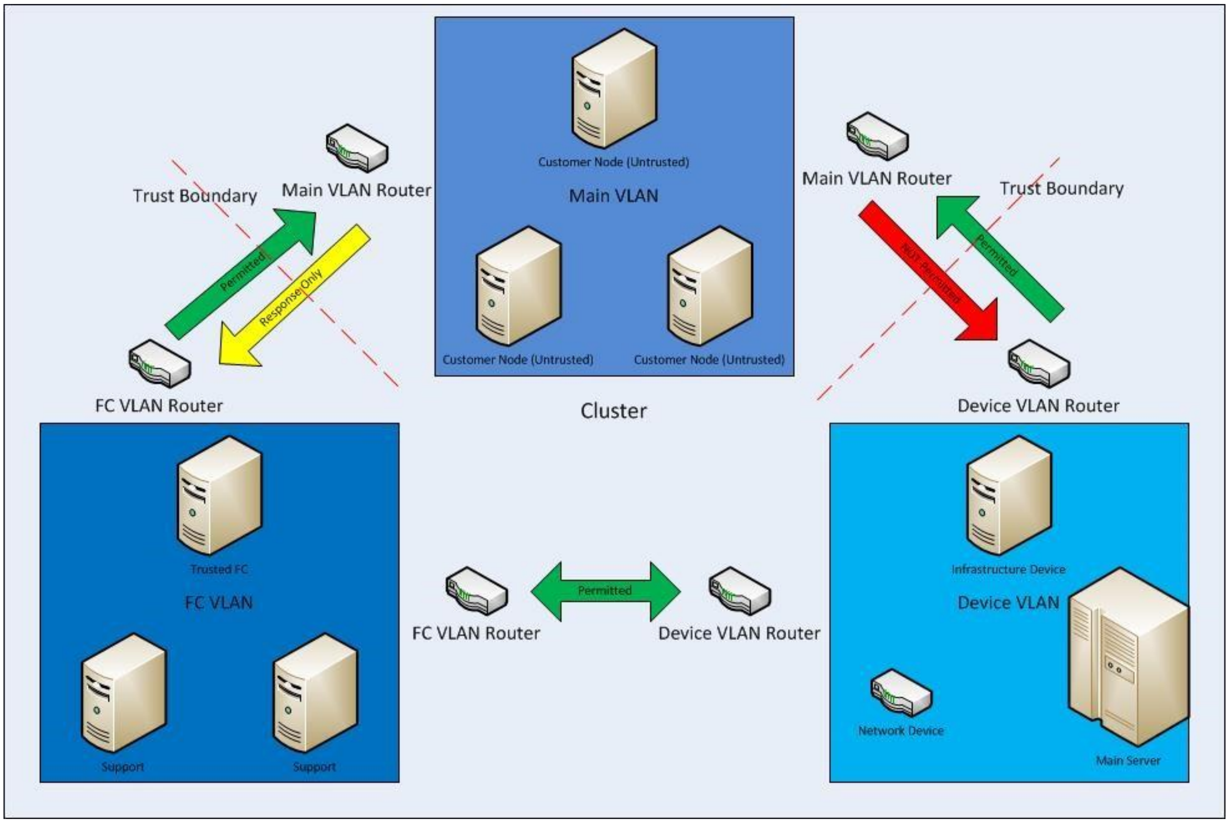
1. [Introduction to Azure Elastic SAN | Microsoft Learn]

VPN/Vlan Concepts

1. Quickstart: Use the Azure portal to create a virtual network - Azure Virtual Network | Microsoft Learn
2. Describe Azure virtual private networks - Training | Microsoft Learn
3. A virtual local area network (VLAN) is a virtualized connection that connects multiple devices and network nodes from different LANs into one logical network.
4. Connect an on-premises network to Azure - Azure Architecture Center | Microsoft Learn
5. Enable remote work by using Azure networking services | Microsoft Learn
6. Azure VNet - An In depth introduction to Azure Virtual Network Monitoring (netreo.com)







Social Coding

GitHub
Repo

Team

Office 365

PowerBI
Services
Reporting

Azure
Resources

stores code
versions code
shares code
built in tasks and schedulers (agents)

Managed - Serverless (no Operating System)

Reason to exist:

Azure Resources

SQL Database - store my order - who it belongs to

Storage - pics/vids (Blobs) file manager

GPS Maps

Secure stuff - users/groups Active Directory

VM - when I need a computer to do stuff
add to network - users -
- maintenance - updates Managed

Convert Video
mp3 - mp4 - ogg

the
app
order &
deliver

app searches
- Search Service

HOST Virtualization Software Hypervisor

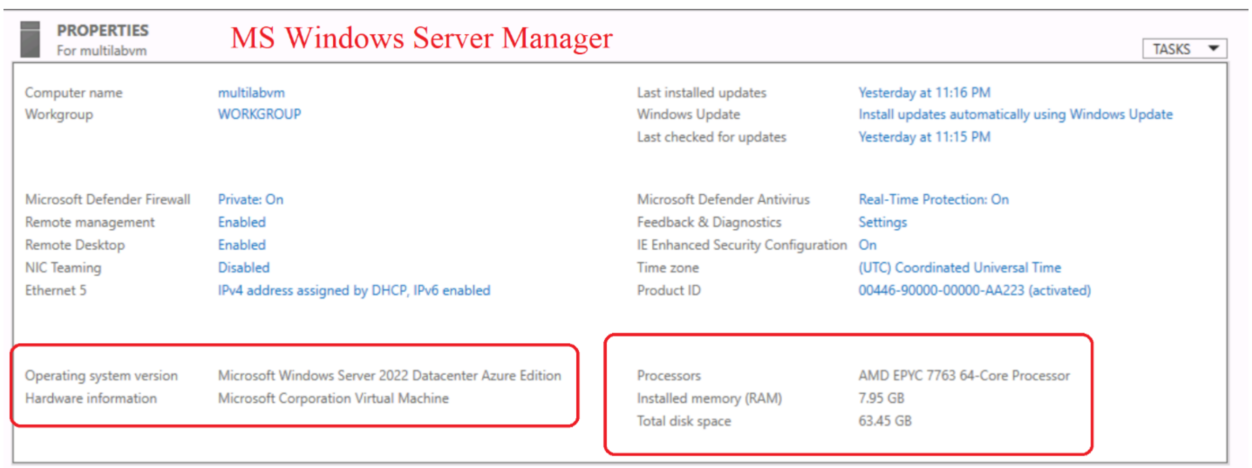
MyPC - Virtual Box or Hyper- v

VM's

- Parasite

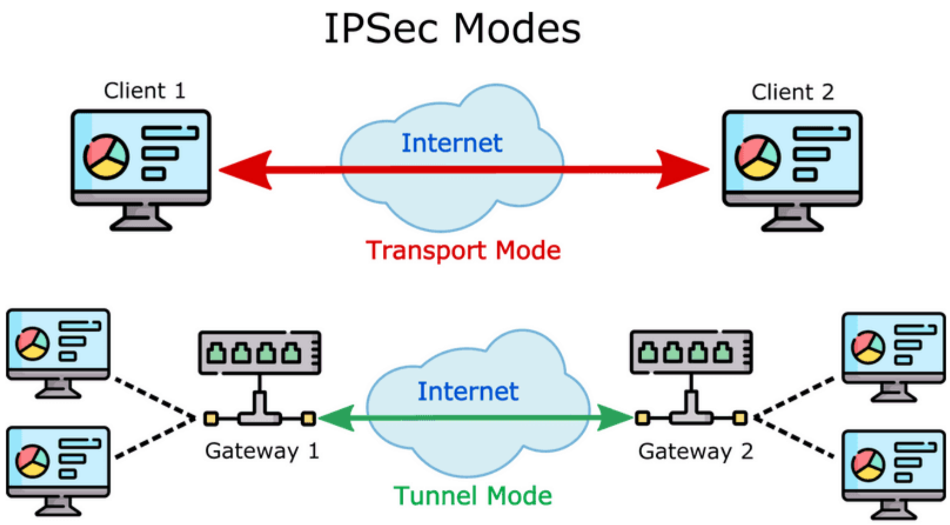
Share Resources
RAM - Hard Drive
Space

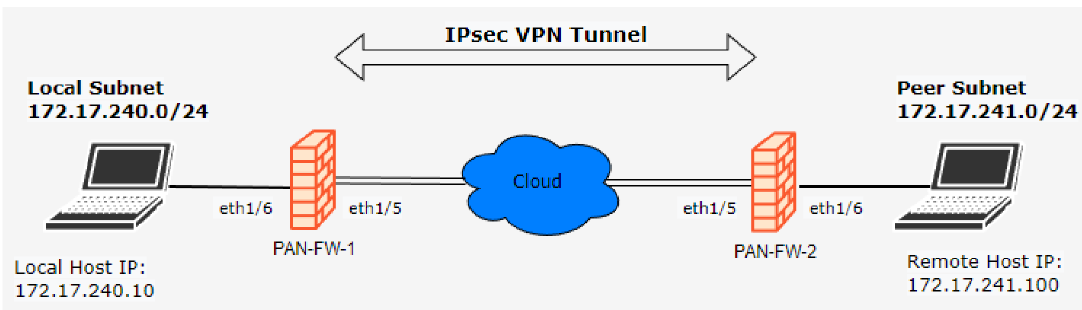
MS Windows Server Manager



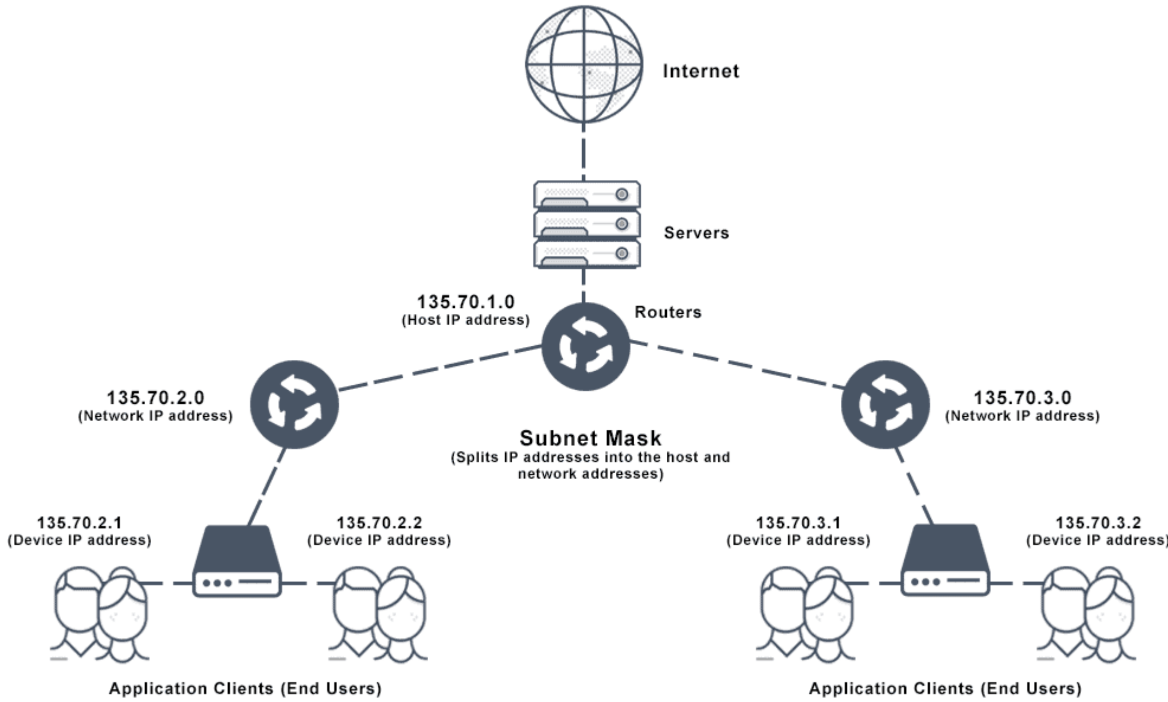
IPsec VPN Tunnel

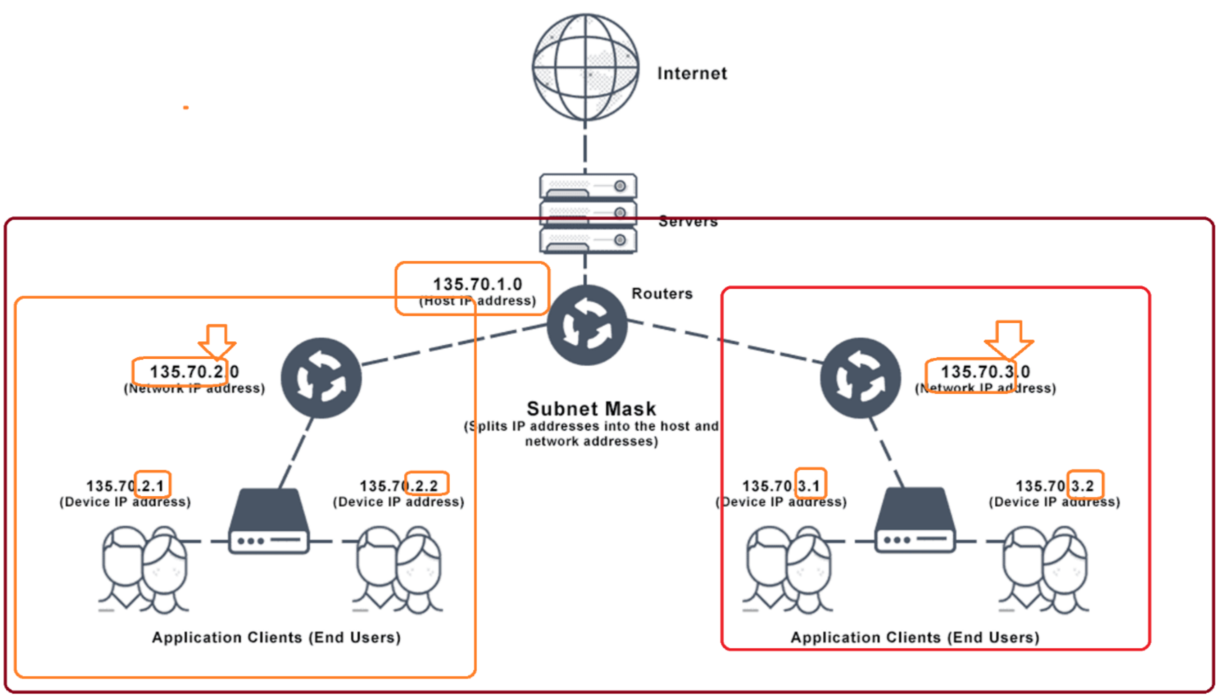
An IPSec tunnel allows for the implementation of a virtual private network (VPN) which an enterprise may use to securely extend its reach beyond its own network to customers, partners, and suppliers.





IP Addressing/Subnets





Subnet Mask Definition

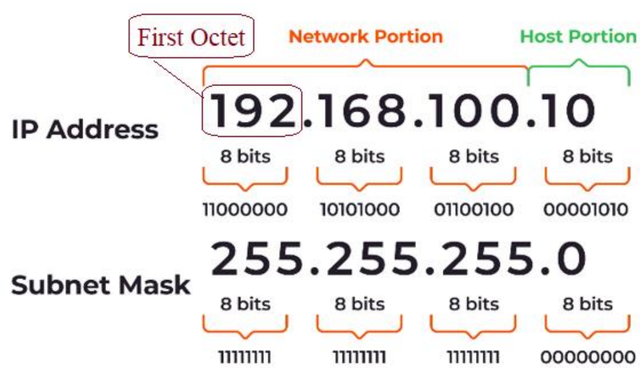
[TCP/IP addressing and subnetting - Windows Client | Microsoft Learn]

Every device has an IP address with two pieces: the client or **host** address and the server or **network** address.

- IP addresses are either configured by a DHCP server or manually configured (static IP addresses).
- The subnet mask splits the IP address into the host and network addresses, thereby defining which part of the IP address belongs to the device and which part belongs to the network.
- IS a four-octet number used to identify the network ID portion of a 32-bit IP address.

The device called a gateway or default gateway connects local devices to other networks.

This means that when a local device wants to send information to a device at an IP address on another network, it first sends its packets to the gateway, which then forwards the data on to its destination outside of the local network.

- The IP address, subnet mask and gateway or router comprise an underlying structure—the Internet Protocol
- The Internet Protocol (IP) is the method for sending data from one computer to another over the internet.
- Each computer, or host, on the internet has at least one IP address as a unique identifier.
- A subnet, or subnetwork, is a network inside a network.
- Subnets make networks more efficient as through subnetting, network traffic can travel a shorter distance without passing through unnecessary routers to reach its destination.
- A subnet mask is a 32-bit number created by
  - Setting **host** bits to all 0s and
  - Setting **network** bits to all 1s
  - 

- This is how the subnet mask separates the IP address into the network and host addresses.
  - The "255" address is always assigned to a broadcast address
  - The "0" address is always assigned to a network address.
  - Neither can be assigned to hosts as they are reserved for these special purposes.
- Subnetworking: Subnetting divides the host element of the IP address further into a subnet.
  - The goal of subnet masks is simply to enable the subnetting process.
  - The phrase "mask" is applied because the subnet mask essentially uses its own 32-bit number to mask the IP address.

What does the subnet mask 255.255 255.0 mean?

For example, a household home network has a standard subnet mask of 255.255. 255.0. This implies using 254 usable IP addresses within the defined network.

One can connect up to 254 internet-enabled devices such as phones, computers, IoT gadgets, and others to the home network to access the internet.

Although the phrase "subnet mask" is preferred, you might use "IP/Mask"

- To define both the IP address and submask at once.
- The IP address is followed by the number of bits in the mask.

  Why is the subnet mask 255.255 255.0 24?

  in binary is

  11111111.11111111. 11111111.00000000

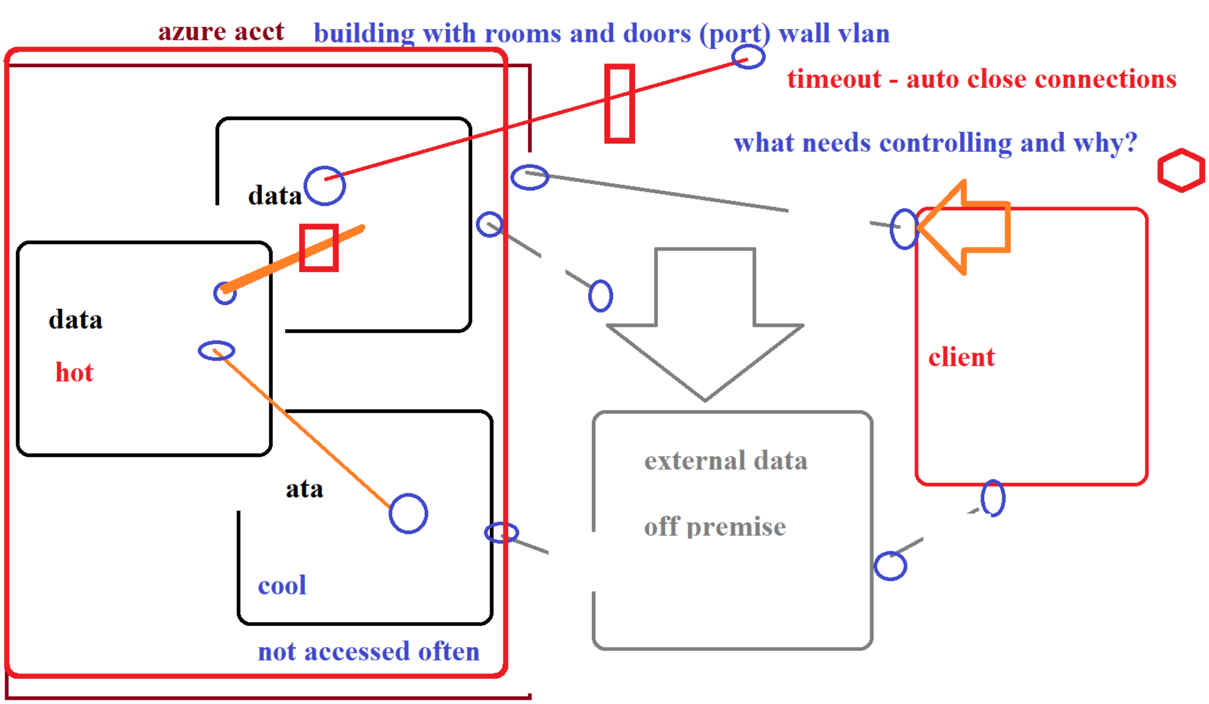
  This adds up to 24 consecutive ones, or /24

  (pronounced "slash twenty four").
-

For example:

- 10.0.1.1/24 is the same as:
  - IP address: 10.0.1.1 with subnet mask of 255.255.255.0
- 216.202.192.66/22
  - IP address: 216.202.196.66 with a subnet mask example of 255.255.252.0

However, you do not mask the IP address, you mask the subnet.



My "Electronic Box"

- An addressing scheme for a range of networks exists based on how the octets in an IP address are broken down.

IP addresses are divided into classes.

1. The most common of them are classes A, B, and C.
2. Each of the address classes has a different default subnet mask.
3. You can identify the class of an IP address by looking at its first octet.

Following are the ranges of Class A, B, and C Internet addresses, each with an example address:

1. Class A networks use a default subnet mask of 255.0.0.0 and have 0-127 as their first octet. The address 10.52.36.11 is a class A address. Its first octet is 10, which is between 1 and 126, inclusive.
2. Class B networks use a default subnet mask of 255.255.0.0 and have 128-191 as their first octet. The address 172.16.52.63 is a class B address. Its first octet is 172, which is between 128 and 191, inclusive.
3. Class C networks use a default subnet mask of 255.255.255.0 and have 192-223 as their first octet. The address 192.168.123.132 is a class C address. Its first octet is 192, which is between 192 and 223, inclusive.
4. EndList

What is the difference between cloud PC and Azure Virtual Desktop?

The difference is that Windows 365 subscriptions reside fully in Microsoft's Azure subscription and are fully managed by Microsoft at fixed costs. Meanwhile, Azure Virtual Desktop (AVD) is entirely customer-managed with flexible consumption-based pricing.

Network Subnets for Kids

Tuesday, TD 🌞 — that's a great idea! Explaining IP addresses and subnets visually — especially "for kids" — is best done like showing **houses on a street** or **students in a classroom**.

Let's make it **super simple and visual**, like a mini story.

🏠 Imagine a Neighborhood (Network)

- The **neighborhood** is like your **network**.
- The **street name** is your **subnet**.
- Each **house** on the street has its **own address** (an IP address).

💡 Example: One Subnet

Let's pick an easy subnet:

192.168.1.0/24

That looks technical, but here's what it means:

Concept	Example	Explanation
Network name (street)	192.168.1.0	The start of the street — identifies the neighborhood.
Subnet mask	/24 = 255.255.255.0	Means there are 256 possible house numbers (from .0 to .255).
Usable IPs (houses)	192.168.1.1 to 192.168.1.254	Each device gets one "house address."
Broadcast address	192.168.1.255	A "megaphone" that shouts to everyone on the street.

🧑‍💻 Example Devices in That Subnet

Device	IP Address	Description
Router (the post office)	192.168.1.1	Delivers data mail to the right house or outside the neighborhood.
Laptop	192.168.1.10	A house on the street.

Device	IP Address	Description
Phone	192.168.1.11	Another house.
Printer	192.168.1.12	Another house that everyone shares.

📊 Simple Diagram (text style)

```
      [ Internet ]
          |
      [ Router: 192.168.1.1 ]
          |
   -------------------------------------
   |      |       |       |
Laptop  Phone  Printer  Smart TV
.10    .11    .12    .13
```

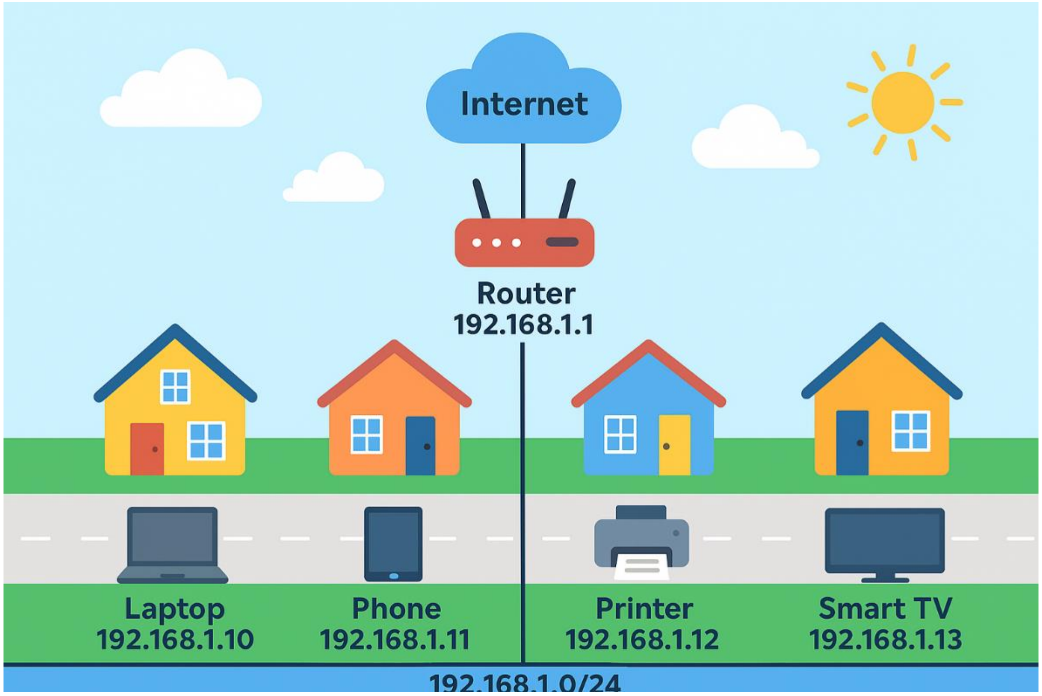
(Each is part of 192.168.1.0/24 subnet)

🧩 Kid-Friendly Summary

The network is like a neighborhood.
The subnet is the street name.
Each device has a house number (IP).
The router is the mail carrier who sends messages between neighborhoods (networks).

Visual Net Neighborhood Diagram

(like a cartoon of a neighborhood with labels: router, devices, subnet line) — would you like me to generate that next?


Internet
Router
192.168.1.1
Laptop
192.168.1.10
Phone
192.168.1.11
Printer
192.168.1.12
Smart TV
192.168.1.13
192.168.1.0/24

Admin Tasks & Dashboard & Notifications & Billing & Scaling & SLA & Service Tiers

1. Create a dashboard in the Azure portal - Azure portal | Microsoft Learn
   a. Create an Azure portal dashboard by using an Azure Resource Manager template - Azure portal | Microsoft Learn
   b. Create an Azure portal dashboard - Code Samples | Microsoft Learn – show site
   c. Create an Azure Monitor dashboard and alerts | Microsoft Learn
2. Create Azure Monitor alert rules - Azure Monitor | Microsoft Learn
   a. Tutorial - Create a metric alert for an Azure resource - Azure Monitor | Microsoft Learn
3. Work with Azure service principals – Azure CLI | Microsoft Learn
4. Create an Azure Policy
5. Open a Support Request
6. Manage access with RBAC
7. Implement resource tagging
8. Manage resource locks
9. Secure network traffic
10. Implement Azure Key Vault
11. Use the Pricing Calculator | Microsoft Azure
12. Create your first function in the Azure portal | Microsoft Learn
13. Azure Resource Graph | Microsoft Azure
14. Cost of Ownership Calculator | Microsoft Azure
15. Templates overview - Azure Resource Manager | Microsoft Learn
16. Azure Network Watcher overview | Microsoft Learn
17. Quickstart for Azure Cloud Shell | Microsoft Learn

Billing in Bullet Points

✅ **Azure Billing Units Overview**

Service	Billing Units / Metrics
Azure SQL Database	- vCore or DTU (**per hour**) - Storage (**GB/month**) - Backup storage (**GB/month**) - IO/s (Hyperscale/Premium tiers)
Virtual Machine (VM)	- Compute (**per second/minute/hour**, based on size) - OS & Data disk storage (**GB/month**) - Network bandwidth (**GB outbound/month**) - Licensing (Windows Server, SQL, etc.)
Virtual Network (VNet)	- Basic features: **Free** - Peering traffic (**GB/month**)

Service	Billing Units / Metrics
	- VPN Gateway (**per hour**) - NAT Gateway (**per hour + GB processed**)
IoT Hub	- Messages (**per million messages/day**) - IoT Hub units (**per hour**) - Device-to-cloud & cloud-to-device ops - Retention & routing (Premium/Standard tiers)
Storage Account	- Blob/Files/Table/Queue storage (**GB/month**) - Transactions (**per 10,000 ops**) - Data transfer (outbound **GB/month**) - Premium SSD vs Standard HDD pricing
App Service	- App Service Plan (**per hour**, based on tier) - Storage (**GB/month**) - SSL certificates (per year)
Azure Functions	- Executions (**per million executions**) - Execution time (**GB-seconds**) - Storage for function app (**GB/month**)
Azure Kubernetes Service (AKS)	- Control plane: **Free** - Node VMs (**per hour**) - Storage (Persistent Volumes **GB/month**) - Network egress (**GB/month**)
Azure Storage (Disks)	- Managed disk size (**GB/month**) - Snapshots (**GB/month**) - Transactions (**per 10,000 ops**)
Azure CDN	- Data transfer (**GB/month**) - Requests (**per 10,000 requests**)
Azure Key Vault	- Operations (**per 10,000 ops**) - Premium tier HSM (**per hour**)
Azure Monitor	- Data ingestion (**GB/month**) - Retention (**GB/month**) - Alerts (**per rule/month**)

✅ Pricing Metrics Quick Reference

- **Compute**: per second/minute/hour depending on service

- **Storage**: per GB/month
- **Network**: outbound data transfer per GB/month
- **Transactions**: per 10,000 operations
- **Executions**: per million executions (Functions, IoT)
- **Retention**: per GB/month for logs/backups

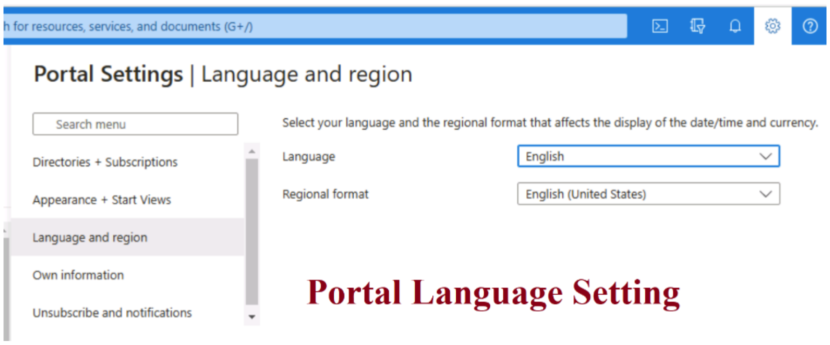
Azure Resource Manager ARM Templates

18. Deploy resources with Azure portal - Azure Resource Manager | Microsoft Learn
19. Deploy and manage resources in Azure by using JSON ARM templates - Training | Microsoft Learn
20. Tutorial - Create and deploy template - Azure Resource Manager | Microsoft Learn
21. Deploy template - Azure portal - Azure Resource Manager | Microsoft Learn
22. When you deploy your ARM template to Azure, go to the Azure portal and make sure you're in the sandbox subscription. To do that, select your avatar in the upper-right corner of the page. Select Switch directory. In the list, choose the Microsoft Learn Sandbox directory.
23. Sample Code from Microsoft Developer Tools | Microsoft Learn

Service Level Agreements

24. Licensing Documents (microsoft.com)
25. Azure SLA Board (azurecharts.com)
26.

Azure CLI & Powershell & Bash & Cmd & Terminal

1. Get started with Azure Command-Line Interface (CLI) | Microsoft Learn
2. *Quickstart for Azure Cloud Shell | Microsoft Learn
    a. Azure CLI interactive mode places you in an interactive shell with autocompletion, command descriptions, and examples.
    b. Azure CLI interactive mode | Microsoft Learn
3. Do you easily mistype in the Windows command prompt as if you were using bash commands, such as "rm" instead of "del"?
    a. Command Prompt Cheatsheet (columbia.edu)
    b. Windows Command Line Cheat Sheet: All You Need in One Place (stationx.net)
4. Compare syntax for Azure Resource Manager templates in JSON and Bicep - Azure Resource Manager | Microsoft Learn
5. 

6. Introduction - Training | Microsoft Learn – Powershell
7. Getting started with PowerShell - PowerShell | Microsoft Learn

🟦 PowerShell

- **Object-oriented**: Works with .NET objects, not just plain text.
- **Rich scripting language**: Supports functions, classes, error handling, etc.
- **Cross-platform** (PowerShell Core): Runs on Windows, Linux, and macOS.
- **Cmdlet-based**: Uses standard verbs like Get-, Set-, New-, Remove-.
- **Integrates deeply with Windows**: Especially for system admin tasks.
- **Can run CMD, Bash, and Azure CLI** commands from within it (with some syntax quirks).

⬛ CMD (Command Prompt)

- **Text-based**: Parses and returns plain text.
- **Windows-only**: Native to Windows OS.
- **Legacy tool**: Limited scripting capabilities (compared to PowerShell).
- **Basic utilities**: dir, copy, echo, etc.
- **Cannot handle complex data structures** like JSON or objects.
- **Not ideal for automation** at scale.

🟦 Azure CLI

- **Cross-platform**: Available for Windows, Linux, macOS.
- **Command-based interface** for managing Azure resources (az).
- **Outputs JSON by default** (easily parseable in scripts).
- **Stateless**: Each command independently interacts with Azure APIs.
- **Designed for Azure automation**: Resource creation, modification, deployment.
- **Can be run inside PowerShell, CMD, or Bash** shells.

🟩 Bash (Bourne Again Shell)

- **Unix/Linux shell**: Default on most *nix systems; available on Windows (via WSL or Git Bash).
- **Text-stream based**: Pipes output between commands as plain text.
- **Powerful scripting**: Loops, conditionals, functions.

- **Huge ecosystem** of command-line tools (awk, sed, grep, etc.).
- **Lightweight & fast**: Great for small scripts and automation.
- **No native object handling**, but supports parsing JSON with tools like jq.

Summary Comparison Table

Feature	PowerShell	CMD	Azure CLI	Bash
Platform	Cross-platform	Windows only	Cross-platform	Cross-platform
Output Type	Objects	Plain Text	JSON	Plain Text
Primary Use	Windows admin	Basic tasks	Azure resource mgmt	Scripting & Unix admin
Scripting Power	High	Low	Moderate	High
Azure Native	No (but usable)	No	Yes	No (needs az)
Data Handling	Object/structured	Text only	JSON, text	Text only

---

Let me know if you'd like examples of commands across all four for comparison.

Absolutely! Here's a side-by-side comparison of **basic shell commands** across **PowerShell**, **CMD**, **Azure CLI**, and **Bash** — all in a clear table format.

Basic Command Comparison Table

Action	PowerShell	CMD	Azure CLI	Bash
List files/directories	Get-ChildItem or ls	dir	*(N/A – not a file shell)*	ls
Change directory	Set-Location or cd	cd	*(N/A)*	cd
Current directory	Get-Location	cd	*(N/A)*	pwd
Create file	New-Item file.txt	echo. > file.txt	*(Use with --output-file)*	touch file.txt
Create folder	New-Item -Type Directory	mkdir	*(Used for Azure resources)*	mkdir

Action	PowerShell	CMD	Azure CLI	Bash
Delete file	Remove-Item file.txt	del file.txt	*(N/A)*	rm file.txt
Delete folder	Remove-Item folder -Recurse	rmdir /S folder	*(N/A)*	rm -r folder
Copy file	Copy-Item file.txt new.txt	copy file.txt new.txt	*(N/A)*	cp file.txt new.txt
Move file	Move-Item file.txt folder/	move file.txt folder\	*(N/A)*	mv file.txt folder/
Read file contents	Get-Content file.txt	type file.txt	*(N/A)*	cat file.txt
Edit file	notepad file.txt	notepad file.txt	*(N/A)*	nano file.txt or vi
Environment vars	$env:PATH	echo %PATH%	*(Used via az config)*	echo $PATH
Print output	Write-Output "Hi"	echo Hi	echo Hi	echo "Hi"
Run Azure login	az login	az login	az login	az login
Azure create resource	*(Use az inside PowerShell)*	*(Use az inside CMD)*	az group create ...	az group create ...

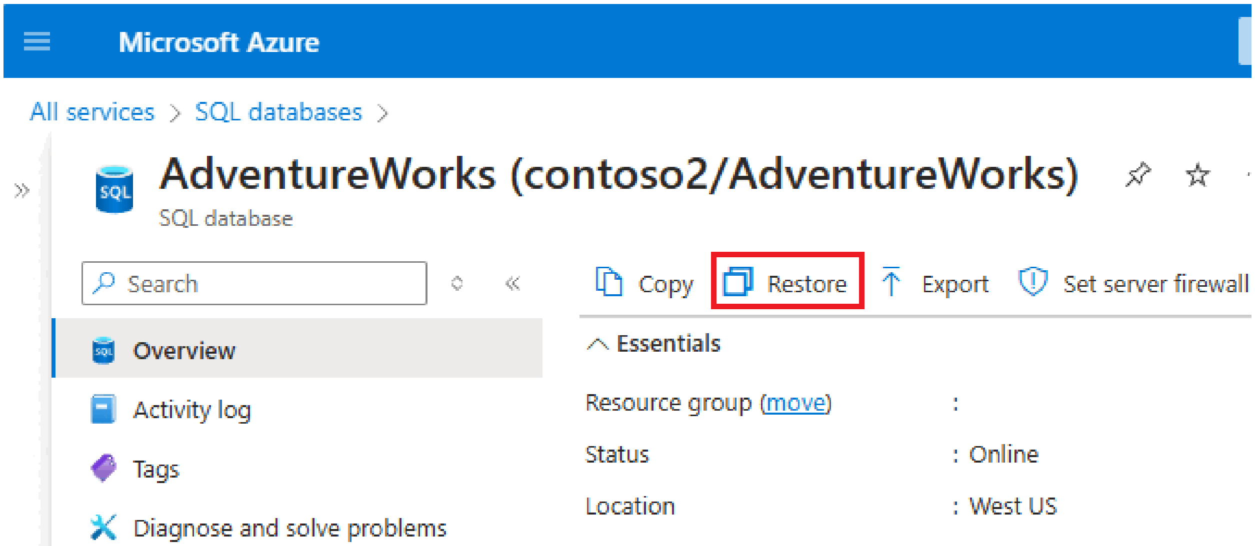
✅ **Notes:**

- **Azure CLI** isn't a full shell — it's used **inside** PowerShell, CMD, or Bash.
- **PowerShell** is verbose but object-oriented.
- **Bash** is concise and great for piping and Unix tooling.
- **CMD** is legacy, basic, and Windows-specific.

Let me know if you'd like a second table for **advanced scripting** or **Azure-specific comparisons**!

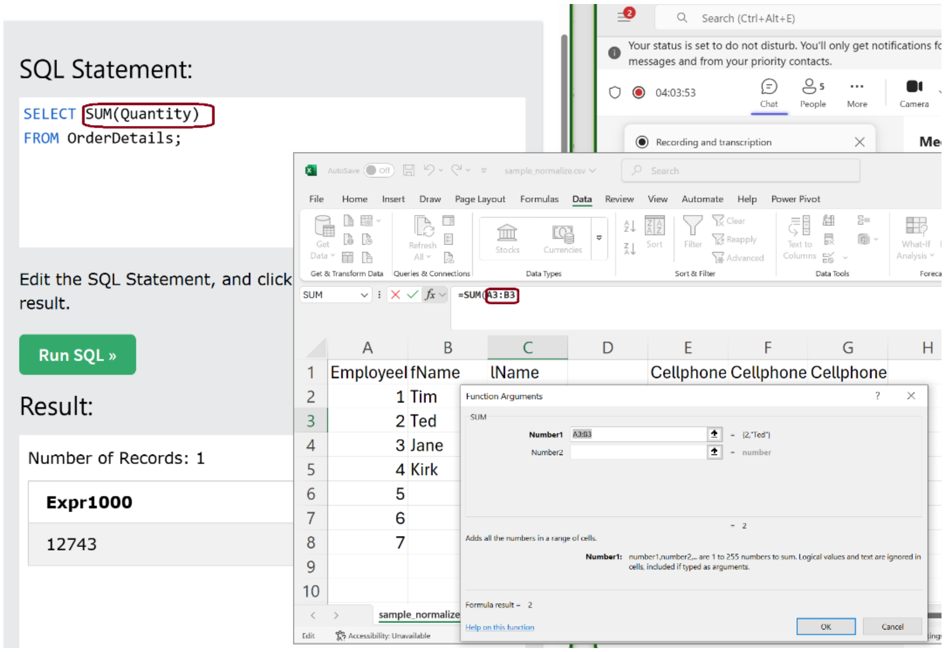
Processes Backup Restore Migrate Import Export Tools SSMS SQL Server Normalize Data Connect

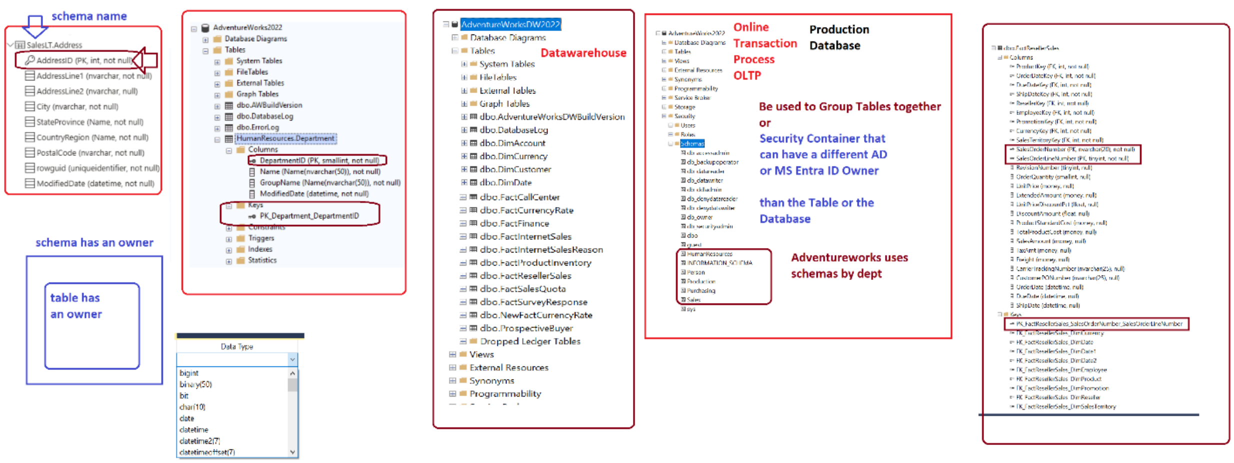
1. Restore a database from a backup - Azure SQL Database | Microsoft Learn
2. The restore list you see in the Azure Portal (like point-in-time restore or deleted database restore) is only populated by the built-in automatic backup system that Azure runs on Azure SQL Database or Managed Instance.
3. To recover a database to a point in time by using the Azure portal, open the database overview page and select **Restore** on the toolbar to open the **Create SQL Database - Restore database** page:

4. 

5. [Database normalization description - Microsoft 365 Apps | Microsoft Learn]()
6. [SQL FOREIGN KEY Constraint]()
7. [Understand star schema and the importance for Power BI - Power BI | Microsoft Learn]()
8. [Create a relationship between tables by using a lookup column - Power Apps | Microsoft Learn]()
9. [Primary and foreign key constraints - SQL Server | Microsoft Learn]()
10. [Relational vs. NoSQL data - .NET | Microsoft Learn]()
11. [Create a database - SQL Server | Microsoft Learn]()
12. [T-SQL Tutorial: Create and query database objects - SQL Server | Microsoft Learn]()

Syntax Code T-SQL Delimiters Common C.R.U.D. Stuff - Editors

13. 

14.
15. 

Azure Data Studio

16. [Connect and Query an Azure SQL Database - Azure Data Studio | Microsoft Learn]
17. [Connect and Query SQL Server Using SSMS | Microsoft Learn]

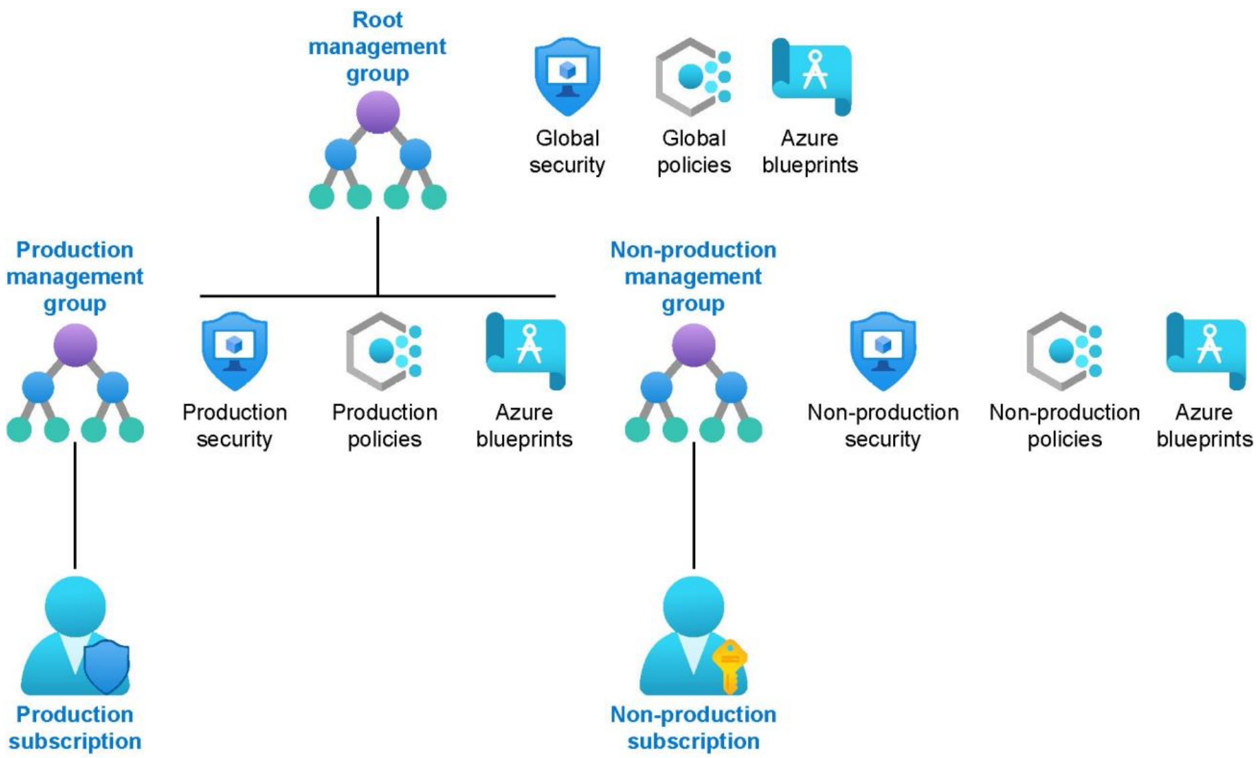
SSMS

18. [Connect and Query Azure SQL Database or Azure SQL Managed Instance | Microsoft Learn]
    a. If you face connectivity issues, check network security groups (NSG) and local firewall/VPN configurations.
    b. Ensure the firewall settings for your Azure SQL server allow connections from your client IP address.

19. SSMS: Connect and query data - Azure SQL Database & Azure SQL Managed Instance | Microsoft Learn
20. Azure Architecture Center - Azure Architecture Center | Microsoft Learn
21. Databases architecture design - Azure Architecture Center | Microsoft Learn

Iot Testing Rasberry Pi

22. Raspberry Pi Azure IoT Web Simulator
23. End List



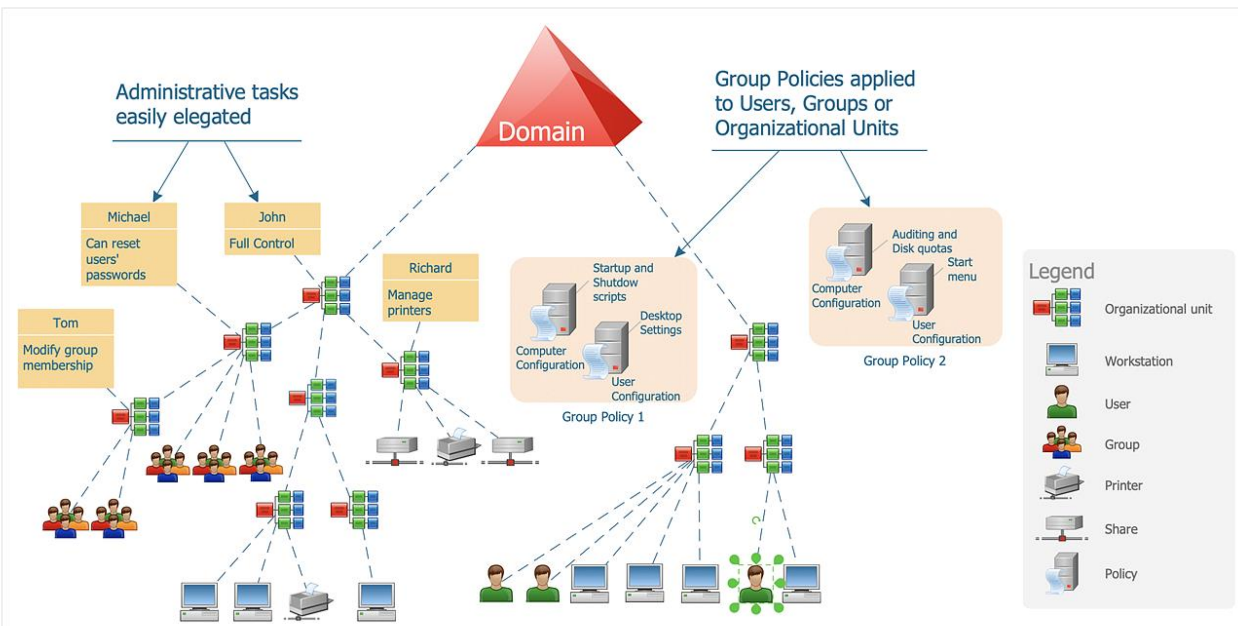
curly brace {} object

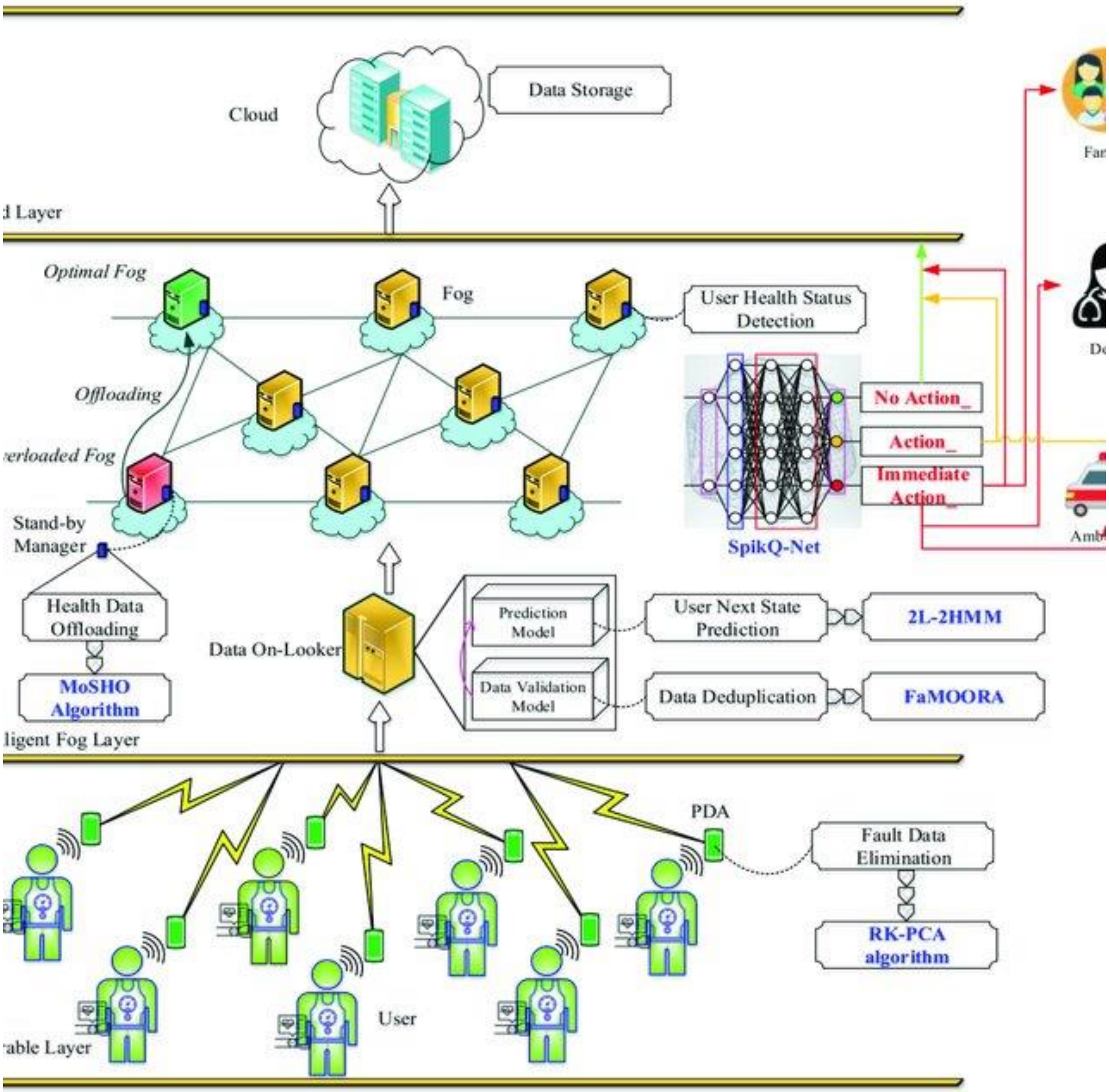
columns:rows

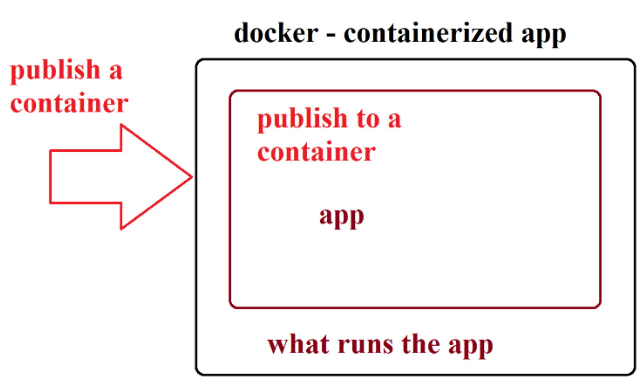
myJSON = '{"name":"John", "age":30, "cars":["Ford", "BMW", "Fiat"]}'

array []

single values like a list of items




Cloud
Data Storage
Optimal Fog
Fog
User Health Status Detection
Offloading
No Action_
Action_
Immediate Action_
Stand-by Manager
SpikQ-Net
Health Data Offloading
Prediction Model
User Next State Prediction
2L-2HMM
Data On-Looker
MoSHO Algorithm
Data Validation Model
Data Deduplication
FaMOORA
PDA
Fault Data Elimination
RK-PCA algorithm
User


docker - containerized app
publish a container
publish to a container
app
what runs the app

Connect to and control other platforms

GitHub
Code
files

Repos -
version
control
in a team
-Security

Azure Arc

connect
to other
systems

Azure Portal

manage
from
portal

containers

DevOps

Info HWY

message - JSON

your money
how does it get
secured?

Point A - Server

Point B
User

gaps/hops

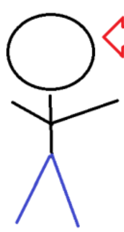
My Data Riverbed

Azure Resources - non Azure Resources

route over the web

watching for something to happen - what is the trigger?

Function - Notify



user access resources - Active Directory

RBAC put users into groups

groups get permissions or policies

control access

public
private

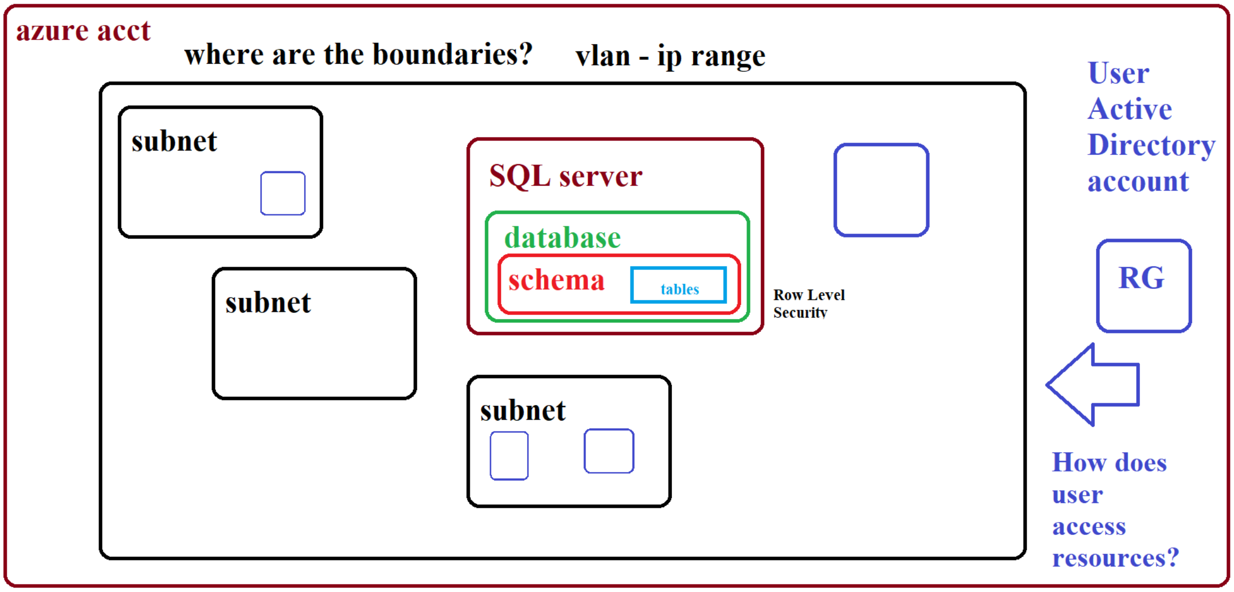
granular - filter or measure

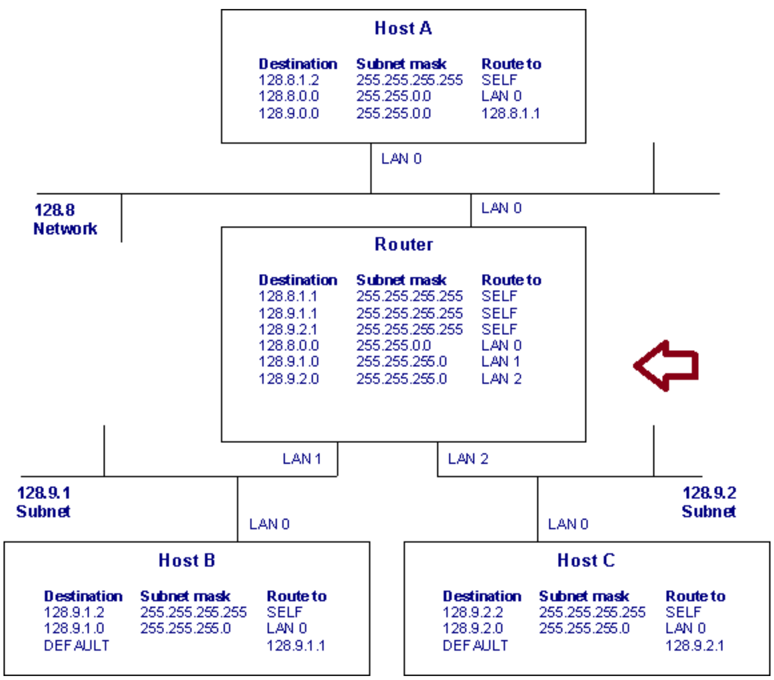
like grains of sand on the beach

bucket

networking pcs together

VLans - group resources


azure acct
where are the boundaries?
vlan - ip range
subnet
subnet
SQL server
database
schema
tables
Row Level Security
subnet
User Active Directory account
RG
How does user access resources?


Host A
Destination Subnet mask Route to
128.8.1.2 255.255.255.255 SELF
128.8.0.0 255.255.0.0 LAN 0
128.9.0.0 255.255.0.0 128.8.1.1
LAN 0
LAN 0
128.8 Network
Router
Destination Subnet mask Route to
128.8.1.1 255.255.255.255 SELF
128.9.1.1 255.255.255.255 SELF
128.9.2.1 255.255.255.255 SELF
128.8.0.0 255.255.0.0 LAN 0
128.9.1.0 255.255.255.0 LAN 1
128.9.2.0 255.255.255.0 LAN 2
LAN 1
LAN 2
128.9.1 Subnet
128.9.2 Subnet
LAN 0
LAN 0
Host B
Destination Subnet mask Route to
128.9.1.2 255.255.255.255 SELF
128.9.1.0 255.255.255.0 LAN 0
DEFAULT 128.9.1.1
Host C
Destination Subnet mask Route to
128.9.2.2 255.255.255.255 SELF
128.9.2.0 255.255.255.0 LAN 0
DEFAULT 128.9.2.1

one pc hard drive - electronic real estate - hosting service

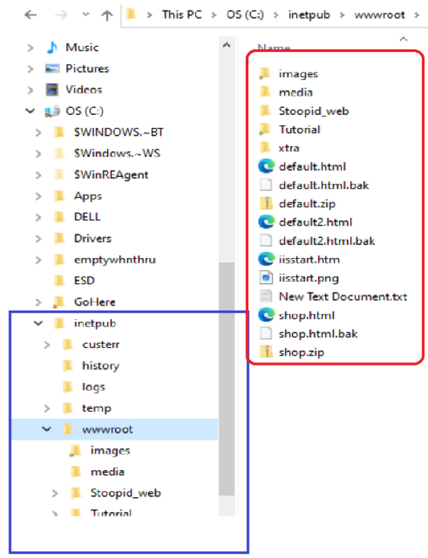
isolated - dedicated - not shared - pay for it

tenant

apt building

Web Server IIS

inetpub
wwwroot
--tenant folders

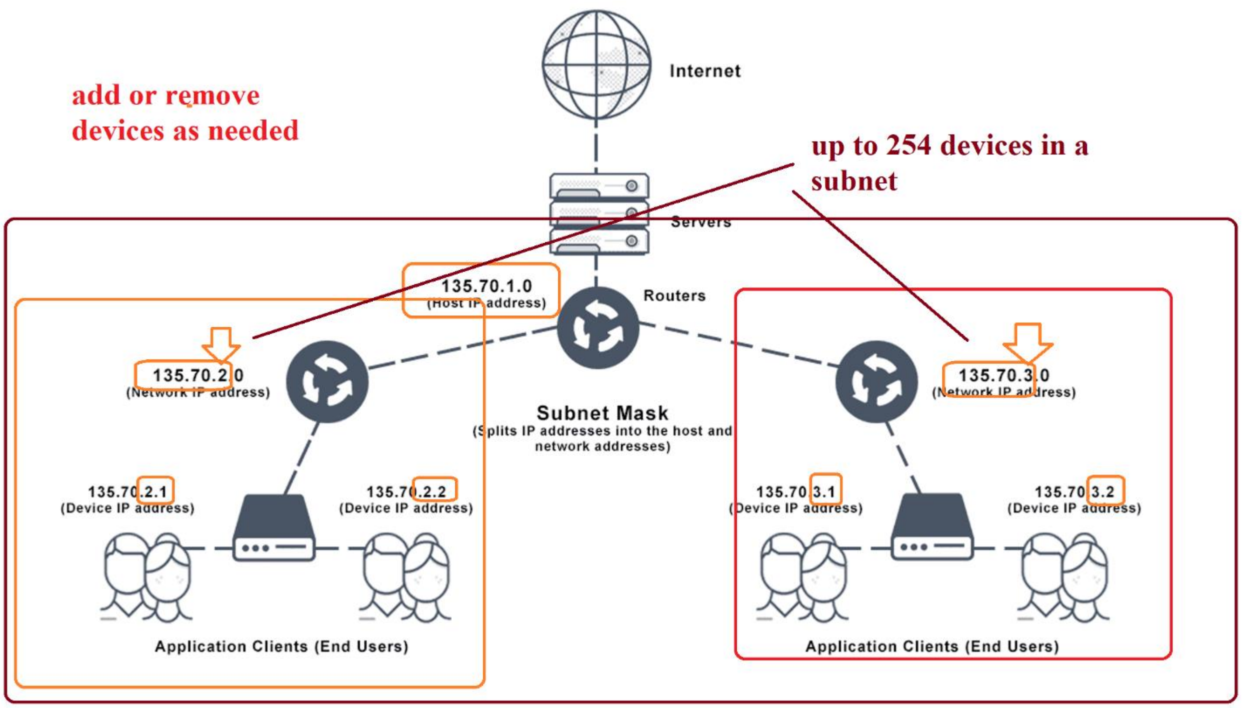


urls: map to record inside IIS
www.myDomain.com

Internet Information Services:
Windows Feature

add or remove devices as needed

up to 254 devices in a subnet



Portals, Icons & Offerings

1. Compliance Offerings Web https://docs.microsoft.com/en-us/compliance/regulatory/offering-home
2. Service Trust Portal: https://servicetrust.microsoft.com/
3. Azure Pricing Calculator webpage: https://azure.microsoft.com/en-us/pricing/calculator/
4. Total Cost of Ownership (TCO) Calculator webpage: https://azure.microsoft.com/en-us/pricing/tco/calculator/
5. SLA summary for Azure services webpage: https://azure.microsoft.com/en-us/support/legal/sla/summary/
6. Azure icons - Azure Architecture Center | Microsoft Learn

7.

Cloud Tech Examples

1. Student Transportation Integrated Suite | Tyler Technologies
2. Trimble SiteVision | High Accuracy Augmented Reality System
3. Workflow: How Uber Works for Drivers and Riders | Overview

Microsoft Learn

1. Study guide for Exam AZ-900: Microsoft Azure Fundamentals | Microsoft Learn
2. Exam AZ-900: Microsoft Azure Fundamentals - Certifications | Microsoft Learn
3. Cloud Model: Blueprints
    a. Pricing Calculator | Microsoft Azure
4. Templates overview - Azure Resource Manager | Microsoft Learn
5. Create a dashboard in the Azure portal - Azure portal | Microsoft Learn
6. Azure information system components and boundaries | Microsoft Learn
7. Azure icons - Azure Architecture Center | Microsoft Learn

More Account Stuff

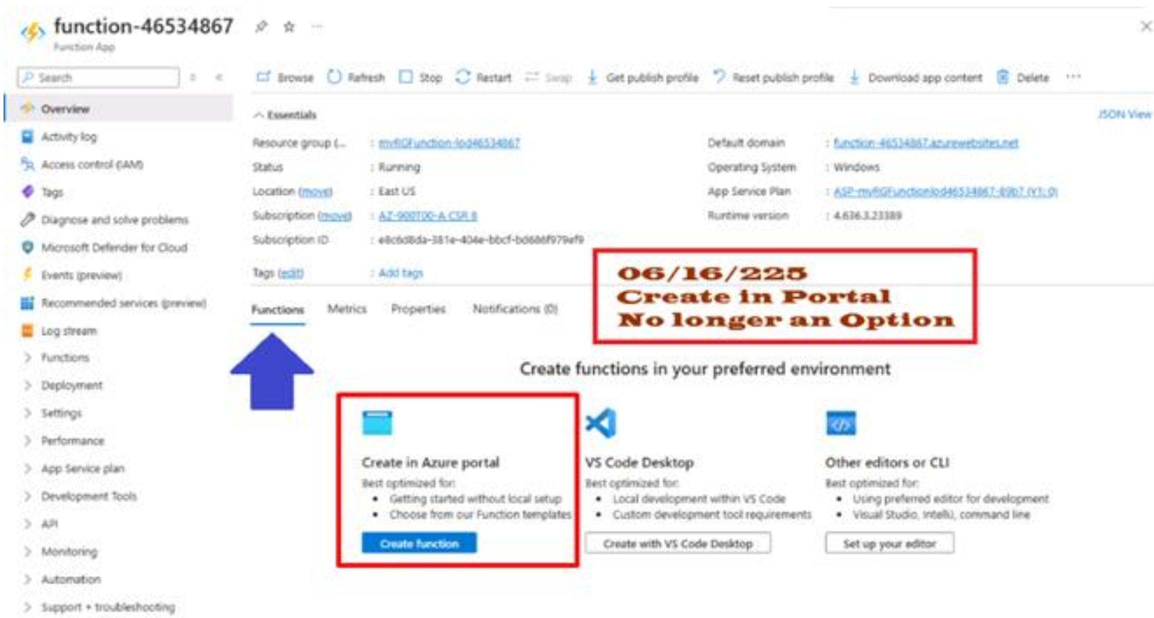
1. Create Your Azure Free Account Today | Microsoft Azure
2. Azure Web PubSub – WebSocket Web Publishing | Microsoft Azure
3. What is Azure Virtual Desktop? - Azure | Microsoft Learn
4. Windows 365 Business Free Trial | Microsoft 365

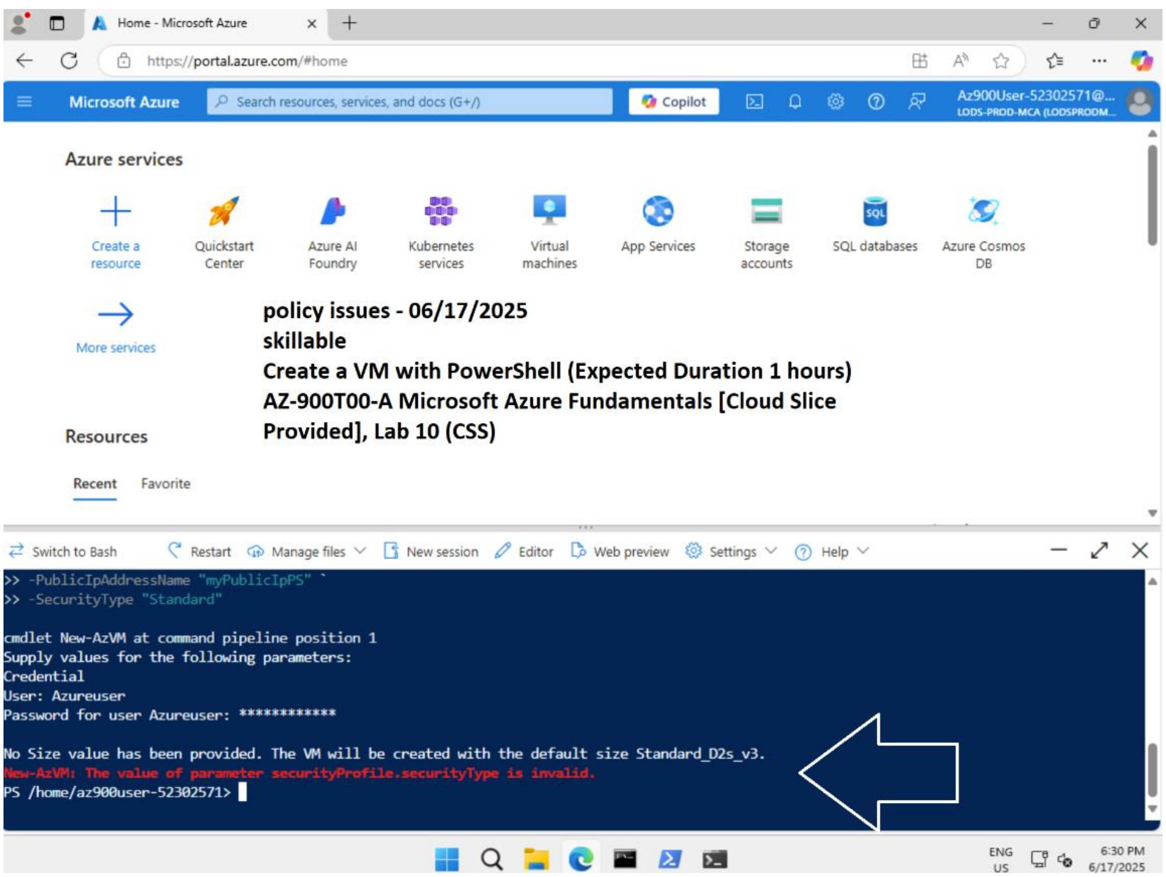
Reference

1. [What is Subnet Mask? Definition & FAQs | Avi Networks]
2. [Manage Azure portal settings and preferences - Azure portal | Microsoft Learn]
3. [Azure Cloud Storage Solutions and Services | Microsoft Azure]
    a. [Persist files in Azure Cloud Shell | Microsoft Learn]
    b. clouddrive command
    c. Enables you to manually update (use) the fileshare that's mounted to Cloud Shell. (like folders on a computer)
    d. Which fileshare is mounted as clouddrive, run the df command.
    e. The file path to clouddrive shows your storage account name and fileshare in the URL.
    f. For example, //storageaccountname.file.core.windows.net/filesharename
    g. Mounting is a process by which a computer's operating system makes files and directories on a storage device (such as hard drive, CD-ROM, or network share) available for users to access via the computer's file system.

End of Doc So Far 2025

Changes in Azure 06/2025





policy issues - 06/17/2025
skillable
Create a VM with PowerShell (Expected Duration 1 hours)
AZ-900T00-A Microsoft Azure Fundamentals [Cloud Slice Provided], Lab 10 (CSS)

▶ Disclaimer (click to expand)

13 Nework Traffic

⚠ Confirm security type is set to **Standard** when creating virtual machines

When creating virtual machines during this lab, we may encounter varying initial defaults for the **Security type** field on Azure Portal. To ensure that virtual machines can be validated and launched, please ensure that the following setting is selected on the **Basics** tab during virtual machine creation:

Settings	Values
Security type	**Standard**

PowerShell **lab 10**

```
New-AzVm `
-ResourceGroupName "myRGPS" `
-Name "myVMPS" `
-Location "East US" `
-VirtualNetworkName "myVnetPS" `
-SubnetName "mySubnetPS" `
-SecurityGroupName "myNSGPS" `
-PublicIpAddressName "myPublicIpPS" `
```

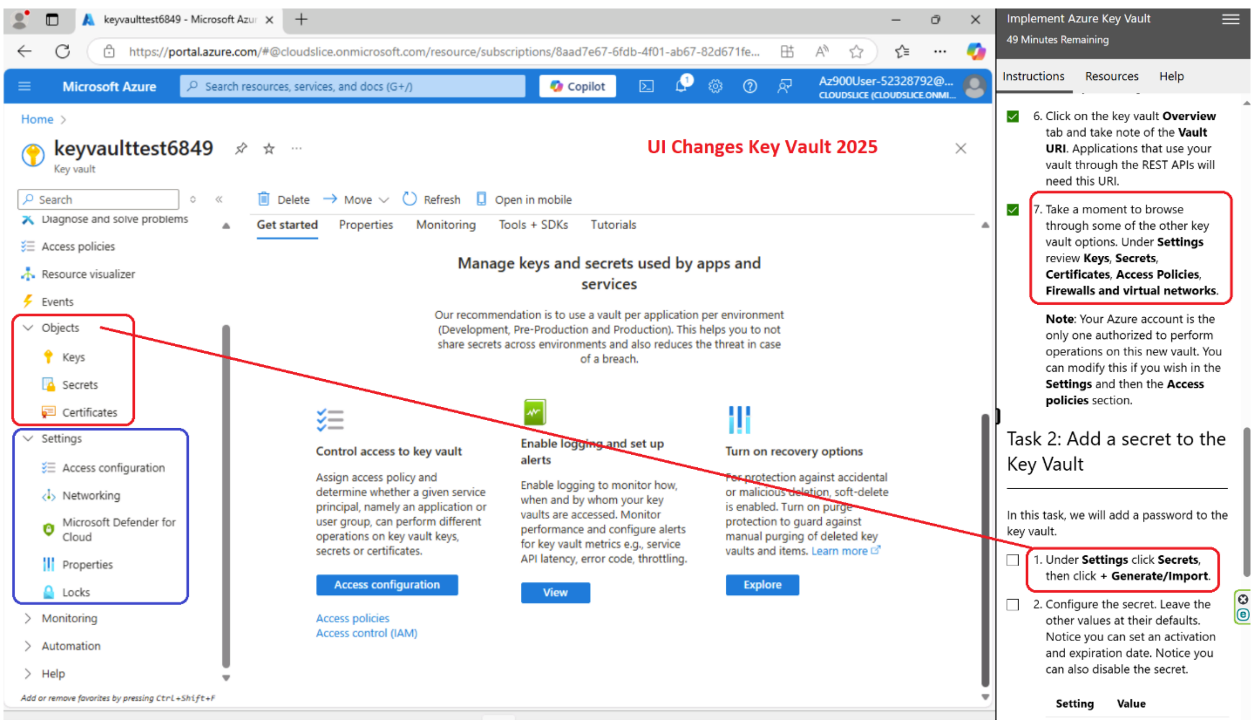
remove live - causes error

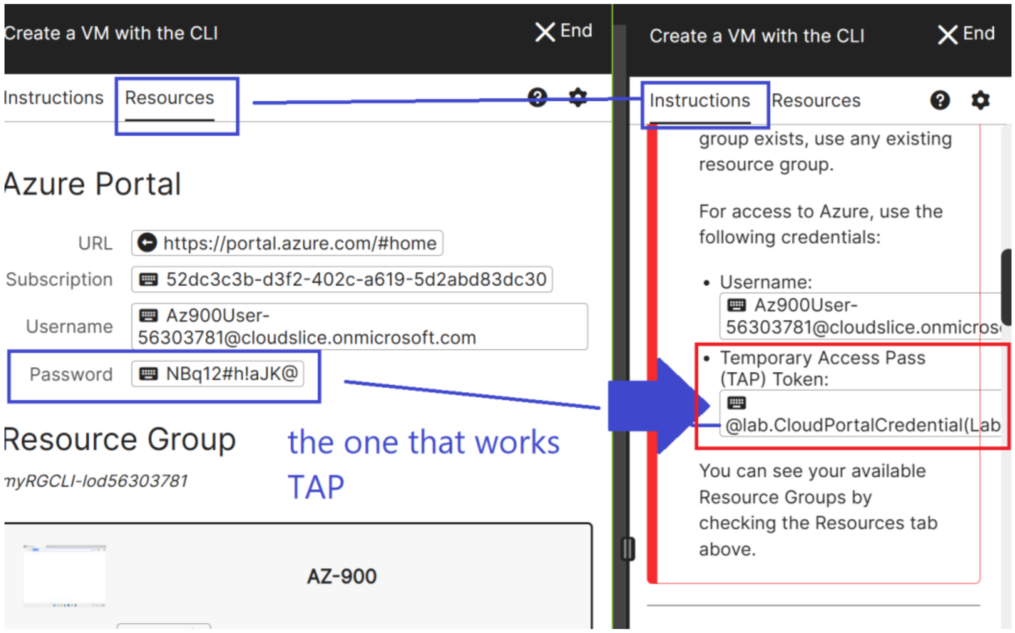
```
-SecurityType "Standard"
```

remove New line

Key Vault lab 12

[Azure Key Vault Overview - Azure Key Vault | Microsoft Learn]




Create a VM with the CLI
End
Instructions
Resources
Azure Portal
URL https://portal.azure.com/#home
Subscription 52dc3c3b-d3f2-402c-a619-5d2abd83dc30
Username Az900User-56303781@cloudslice.onmicrosoft.com
Password NBq12#h!aJK@
Resource Group
myRGCLI-lod56303781
the one that works
TAP
AZ-900
Create a VM with the CLI
End
Instructions
Resources
group exists, use any existing resource group.
For access to Azure, use the following credentials:
Username:
Az900User-56303781@cloudslice.onmicros
Temporary Access Pass (TAP) Token:
@lab.CloudPortalCredential(Lab
You can see your available Resource Groups by checking the Resources tab above.

Create a VM with the CLI ✕ End

Instructions Resources

Azure Portal

URL https://portal.azure.com/#home

Subscription 60dc3063-135a-4ebf-9819-89b638c936a8

Username Az900User-56303921@cloudslice.onmicrosoft.com

Password kPbH7r+8S$m0

TAP J5LWM&BY

Resource Group

myRGCLI-lod56303921

AZ-900

Username Admin

Password Pa55w.rd

Ctrl+Alt+Delete

Open in New Window

Create a VM with the CLI ✕ End

Instructions Resources

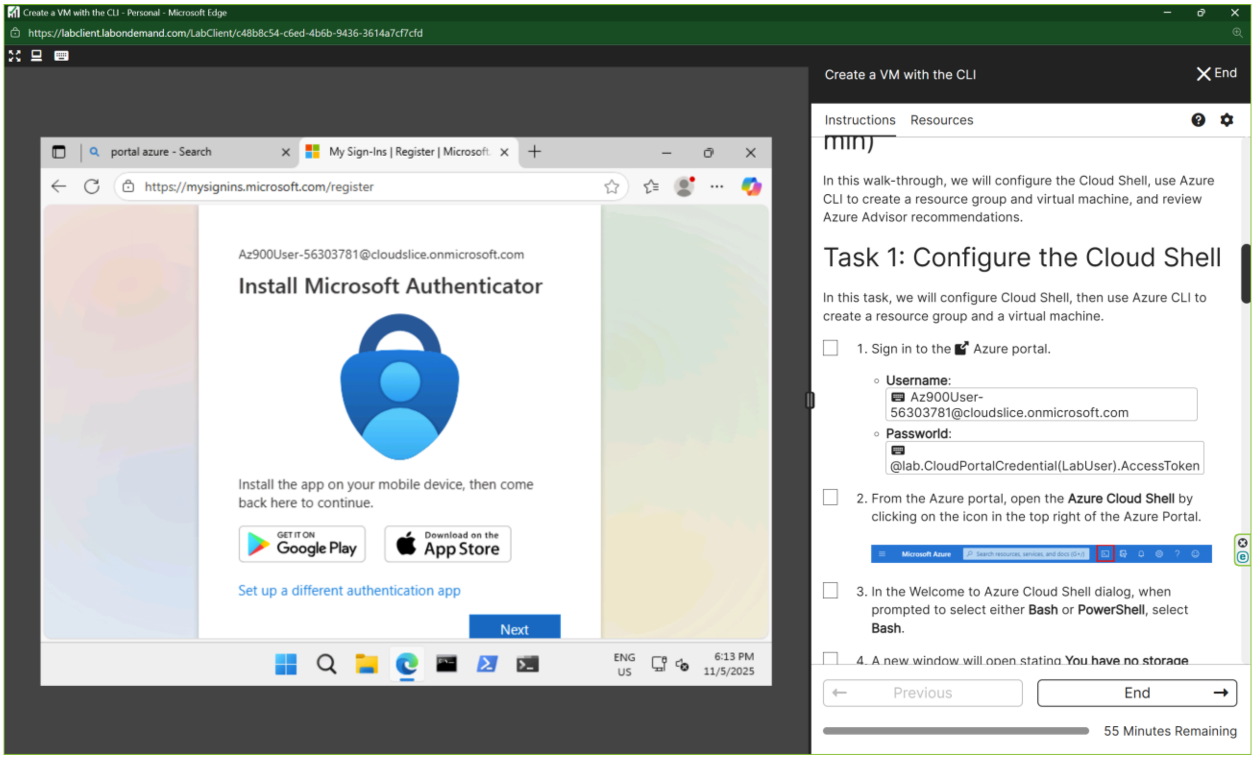
lab setup. You can safely ignore any steps that ask you to create a resource group. Please use or select a pre-existing resource group that has a similar name to the one you are asked to create. If no similarly named resource group exists, use any existing resource group.

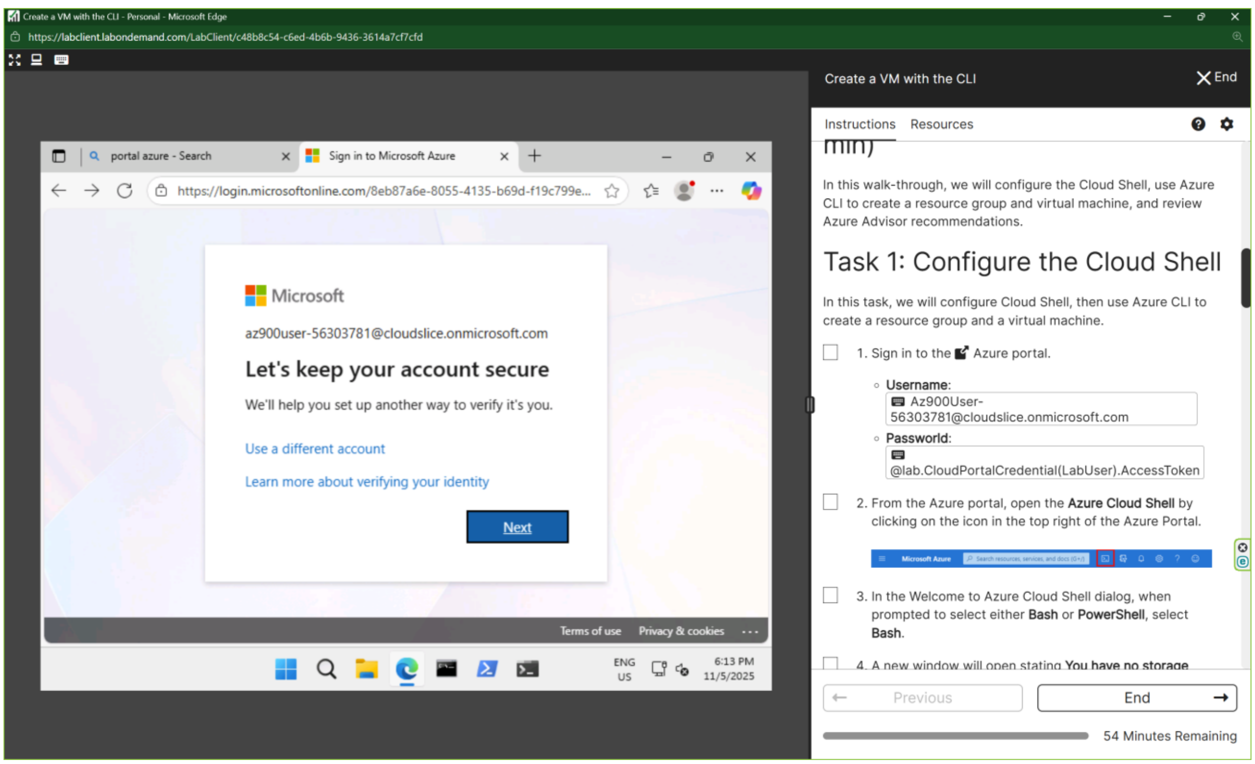
For access to Azure, use the following credentials:

- Username: Az900User-56303921@cloudslice.onmicros
- Temporary Access Pass (TAP) Token: J5LWM&BY

You can see your available Resource Groups by checking the Resources tab above.

11 - Create a VM with the CLI (10 min)


Create a VM with the CLI - Personal - Microsoft Edge
https://labclient.labondemand.com/LabClient/c48b8c54-c6ed-4b6b-9436-3614a7cf7cfd
portal azure - Search
My Sign-Ins | Register | Microsoft
https://mysignins.microsoft.com/register
Az900User-56303781@cloudslice.onmicrosoft.com
Install Microsoft Authenticator
Install the app on your mobile device, then come back here to continue.
GET IT ON Google Play
Download on the App Store
Set up a different authentication app
Next
ENG US
6:13 PM 11/5/2025
Create a VM with the CLI
End
Instructions
Resources
In this walk-through, we will configure the Cloud Shell, use Azure CLI to create a resource group and virtual machine, and review Azure Advisor recommendations.
Task 1: Configure the Cloud Shell
In this task, we will configure Cloud Shell, then use Azure CLI to create a resource group and a virtual machine.
1. Sign in to the Azure portal.
Username:
Az900User-56303781@cloudslice.onmicrosoft.com
Password:
@lab.CloudPortalCredential(LabUser).AccessToken
2. From the Azure portal, open the Azure Cloud Shell by clicking on the icon in the top right of the Azure Portal.
3. In the Welcome to Azure Cloud Shell dialog, when prompted to select either Bash or PowerShell, select Bash.
Previous
End
55 Minutes Remaining


Create a VM with the CLI - Personal - Microsoft Edge
https://labclient.labondemand.com/LabClient/c48b8c54-c6ed-4b6b-9436-3614a7cf7cfd
portal azure - Search
Sign in to Microsoft Azure
https://login.microsoftonline.com/8eb87a6e-8055-4135-b69d-f19c799e...
Microsoft
az900user-56303781@cloudslice.onmicrosoft.com
Let's keep your account secure
We'll help you set up another way to verify it's you.
Use a different account
Learn more about verifying your identity
Next
Terms of use
Privacy & cookies
ENG US
6:13 PM 11/5/2025
Create a VM with the CLI
End
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Previous
End
54 Minutes Remaining